

SENSOR TRANSPORTATION APPARATUS AND GUIDE DEVICE

Technical Field

This invention relates to apparatus and methods for use in transporting sensor equipment, and
5 in particular to apparatus and methods for use in wireline logging applications.

Background art

Hydrocarbon exploration and development activities rely on information derived from sensors which capture data relating to the geological properties of an area under exploration. One
10 approach used to acquire this data is through wireline logging. Wireline logging is typically performed in a wellbore immediately after a new section of hole has been drilled. These wellbores are drilled to a target depth covering a zone of interest, typically between 1000 - 5000 meters deep. A sensor package, also known as a "logging tool" or "tool-string" is then lowered into the wellbore and descends under gravity to the target depth of the wellbore well. The
15 logging tool is lowered on a wireline – being a collection of electrical communication wires which are sheathed in a steel cable connected to the logging tool. Once the logging tool reaches the target depth it is then drawn back up through the wellbore at a controlled rate of ascent, with the sensors in the logging tool operating to generate and capture geological data.

20 There is a wide range of logging tools which are designed to measure various physical properties of the rocks and fluids contained within the rocks. The logging tools include transducers and sensors to measure properties such as electrical resistance, gamma-ray density, speed of sound and so forth. The individual logging tools are often combinable and are typically connected together to form a logging tool-string. These instruments are relatively
25 specialised sensors, which in some cases need to be electrically isolated or located remote from metallic objects which are a source of noise in the data generated. Some sensors are designed to make close contact with the borehole wall during data acquisition whilst others are ideally centred in the wellbore for optimal results. These requirements need to be accommodated with any device that is attached to the tool-string.

30 The drilling of wells and the wireline logging operation is an expensive undertaking. This is primarily due to the capital costs of the drilling equipment and the specialised nature of the wireline logging systems. It is important for these activities to be undertaken and completed as promptly as possible to minimise these costs. Delays in deploying a wireline logging tool are to
35 be avoided wherever possible.

One cause of such delays is the difficulties in lowering wireline logging tools down to the target depth of the wellbore. As the logging tool is lowered by cable down the wellbore by gravity alone, an operator at the top of the well has very little control of the descent of the logging tool.

5 Logging tools can become held up on ledges of rock formed inside a well – usually found at the boundaries of hard rock layers where the adjacent rock layer crumbles. An operator may not immediately identify that a logging tool has become stuck on a ledge, and may also spend a significant amount of time reeling the cable and tool-string back in and attempting to move it past the obstruction formed by a ledge.

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The chances of a wireline logging tools getting stuck or being impeded is also significantly increased with deviated wells. Deviated wells do not run straight vertically downwards and instead extend downward at an angle. Multiple deviated wells are usually drilled from a single surface location to allow a large area of interest to be explored. As wireline logging tools are run
15 down a wellbore with a cable under the action of gravity, the tool-string will traverse the low side or bottom of the wellbore wall and immediately encounter any obstructions on the wellbore wall as it travels downwards to the target depth.

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Furthermore, in deviated wells there is the potential for drilling cuttings to collect on the low side of the wellbore. Rock cuttings are more difficult to remove when the wellbore is deviated. The logging tool has to travel over or through these drilling cuttings, which can impede its progress and also collect in front of the logging tool. In some cases the logging tool may not be able to plough through the cuttings to reach the bottom of the wellbore.

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Furthermore, as hole deviation increases, the sliding friction can prevent the logging tool descending. The practical limit is around 50-60° from the vertical, and in these high angle wells any device that can reduce friction is very valuable. The running of the tool-string over the low side surface of the borehole also needs to be taken into account in the design of the tool-string, and in particular the housing and the containment of its sensitive sensors and transducers.

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Attempts have been made to address these issues in the deployment of wireline logging tools, as disclosed in US patents 7866384, 7395881 and US patent application 20120061098. These patent specifications describe a number of different forms of an in-line roller devices integrated into the logging tool-string. These devices aim to reduce the friction experienced by a tool-string
35 as it is run along the low side slope of a deviated wellbore.

This in-line positioning of roller devices increases the potential to cause damage to the logging tools as there are additional O-rings connections required that could potentially leak.

Furthermore, there are multiple additional electrical connections that need to be made between

5 the individual logging tools making up the logging tool-string. The additional components and tool-string length means it takes longer to connect and disconnect the tool-string, which slows down the wireline logging operation and therefore increases well costs. Further, there is a lack of flexibility inherent in this approach as in-line roller devices can only be placed between logging tools, and as some of these tools are quite long, in-line rollers may not keep the entire
10 tool body off the borehole wall.

The rollers employed in these forms of prior art devices also have relatively small wheels with a minimal clearances. In deviated wellbores well drilling cuttings will collect on the low side of the well, and these small sets of rollers can struggle to make headway through piles of cuttings. In

15 situations where large amounts of cuttings are encountered these small rollers can be of no assistance at all and simply add to the length and weight of an already unwieldy tool.

McNay patent US8011429 and Schumberger patent application US2013248208 describe roller assemblies which slip over the logging tool, and are mounted such that they are free to rotate

20 about the longitudinal axis of the tool. These devices have relatively small wheels which do not rotate easily over rough surfaces. In addition, it is often the central or side part of the wheel which is in contact with the wellbore wall, rather than the circumferential or radially extreme edge. This means that the wheels are often skidding rather than rotating. Neither of these devices has an active lubrication system which can prevent contaminants from entering and
25 jamming the bearings.

All of these existing prior art devices attempt to assist a tool-string in travelling down a deviated wellbore. However, the devices do not assist in maintaining a known orientation of the tool-string, or a specific clearance or “standoff” between the active part of the tool-string and the
30 wellbore wall.

US 684732 describes a sectional threading rod which is provided with wheels to reduce friction as the rod is pushed down a conduit. The wheels have an axle which is above the centre of the rod. However, the rod is not intended for use with a logging tool

Other attempts to address the issues associated with deviated wellbores include a number of prior art "hole finding" devices. For example US patent US4474235, US patent application US 20120061098, PCT publication WO 2010/106312 and US patent application US20120222857 all describe systems for wireline hole finding devices which rely on one or more rollers located at the nose at the bottom of a tool-string. The nose is the leading end at the bottom of the tool-string during descent of the wellbore. These rollers are arranged to allow the nose of the tool-string to roll into, and then up and over, irregularities and obstructions in a wellbore.

However due to the use of a number of metallic components these types of hole finding devices are not necessarily compatible with induction type resistivity tools which are generally the most commonly used sensor in such applications. Also, some logging tools are "bottom only", and have sensors which must be located at the lower extremity of the tool-string. The prior art hole finders are heavy and metallic and hence not compatible with such tools.

Furthermore, these systems are relatively complicated and must be appropriately designed and maintained to withstand the hostile environment experienced at depth in exploration wells. If the moving parts used in these systems cease to function the hole finder is ineffective. These designs are also relatively heavy and inflexible. Any impact forces or torque acting on the hole finder are transmitted into the tool-string, potentially causing damage to the sensors.

Another approach used in the design of hole finding devices is disclosed in US patent application US 20090145596. This patent specification describes an alternative hole finding system employed outside of wireline applications where a conduit, tubing or pipe is attached to the sensor tool in order to push it down the wellbore. This specification discloses a relatively complicated system which requires a surface operator to actively adjust the orientation of a nose assembly mounted at the bottom of the tool. The specification also discloses that this device requires a range of sensors that are used to detect sensor tool movement, and specifically if the sensor tool is held up. This form of hole finding system is again relatively heavy and complex. Furthermore, a dedicated operator is also required to monitor the progress of the sensor tool to actively adjust the orientation and angle of attack of the adjustable nose assembly when the sensors detect that the sensor device is held up as it moves down the wellbore. A similar design is described in US patent US7757782. This device is also an active system which requires manipulation from an operator at the surface to change the nose angle and azimuth in order in order to navigate past obstacles after the logging tool is obstructed.

It would therefore be of advantage to have an improved guide device which addressed any or all of the above issues. In particular, it would be of advantage to have an improved guide device which avoided obstacles – as opposed to having to negotiate obstacles. It would also be of advantage to have an improved guide device that did not require monitoring and active manipulation as the logging tool descends the wellbore. An improved guide device formed from a minimum number of metallic or conductive components, which includes no moving parts, which is easy to maintain and manufacture and which is lightweight and simple would be of advantage over the prior art. Furthermore it would also be of advantage to have an improved guide device which, if lost in an exploration well, could be drilled through to remove it as an obstruction.

It would also be of advantage to have an improved sensor transportation apparatus which addressed any or all of the above issues or at least provided an alternative choice. In particular, a sensor transportation apparatus which reduced the friction experienced by a wireline logging tool during deployment down a deviated well would be of advantage. An improved sensor transportation apparatus which could also be used to orient a tool-string would also be of advantage. Furthermore it would also be of advantage to have improvements which coped with the build-up or collection of drilling cuttings in deviated wells and/or which addressed the problems inherent in prior art in-line roller systems.

The reference to any prior art in the specification is not, and should not be taken as, an acknowledgement or any form of suggestion that the prior art forms part of the common general knowledge in any country.

Disclosure of Invention

According to one aspect of the present invention there is provided a sensor transportation apparatus to convey an elongate sensor assembly through a wellbore, the sensor transportation apparatus comprising:

- at least one engagement structure to connect the sensor transportation apparatus to the sensor assembly, and
- one or more wheels arranged to rotate about an axis of rotation substantially perpendicular to a longitudinal axis of the sensor assembly when the transportation apparatus is connected to the sensor assembly, and

an orientation structure defining a form having a transverse outline which has a rotational centre, wherein the rotational centre is offset from a centroidal axis of the elongate sensor assembly.

5 Preferably the lateral extremities of the orientation structure substantially lie on a substantially circular imaginary curve which is centred at the rotational centre.

Preferably the sensor transportation apparatus has only one stable orientation.

Preferably the wheels are shaped and dimensioned such that, in use, contact between each wheel and the wellbore wall is at a surface of the wheel which is at the radial extremity of the wheel, relative to the axis of rotation.

10 Preferably each said wheel extends below the sensor assembly and the engagement structure when the sensor transportation apparatus is in a substantially horizontal orientation.

Preferably a rotational axis of each said wheel is above the centroidal axis of the tool-string when the sensor transportation apparatus is in a substantially horizontal orientation.

Preferably the rotational axes of the wheels are substantially parallel and coplanar.

15 Preferably at least two of the wheels have a common axis of rotation.

Preferably the wheels are mounted on stub axles.

Preferably the orientation structure comprises at least one laterally extending orientation projection.

Preferably the orientation structure further comprises at least one of the wheels.

20 Preferably the wheels run on bearings and the apparatus further comprises a lubrication delivery apparatus which provides a lubricant to the bearings at a pressure which is greater than ambient wellbore pressure.

Preferably the engagement structure engages an exterior surface of the sensor assembly.

25 Preferably the engagement structure partially encloses the exterior surface of the sensor assembly without completely encircling the sensor assembly.

Preferably a portion of the sensor assembly extends below the engagement structure when the sensor transportation apparatus is in a substantially horizontal orientation.

Preferably the engagement structure is adapted for in-line connection to the sensor assembly.

30 Preferably the transportation apparatus further comprises a drive means, wherein at least one of the wheels is connectable to the drive means.

Preferably the apparatus further comprises a clutch between the drive means and each driven wheel.

Preferably the apparatus further comprises a jockey wheel mounted to adjustable mounting means for selectively raising or lowering the jockey wheel.

- 5 Preferably when the mounting means are raised, the jockey wheel engages an opposite side of the wellbore to the wheels.

According to a second aspect of the present invention there is provided a sensor transportation apparatus to convey an elongate sensor assembly through a wellbore, the sensor transportation apparatus comprising

- 10 at least one engagement structure to connect the sensor transportation apparatus to the sensor assembly, and
one or more wheels arranged to rotate on bearings about an axis of rotation substantially perpendicular to a longitudinal axis of the sensor assembly when the transportation apparatus is connected to the sensor assembly, and a lubrication delivery apparatus
15 which provides a lubricant to at least one of the bearings at a pressure which is maintained at greater than ambient wellbore pressure.

Preferably the bearing comprises a ball bearing assembly or a bush bearing assembly.

- 20 Preferably the lubrication delivery apparatus comprises a reservoir comprising an elastic diaphragm.

Preferably the reservoir comprises a volume between an inner surface of the reservoir and a base structure.

Preferably the base structure comprises an outer surface of the transportation means.

Preferably an exterior side of the diaphragm is subject to ambient pressure.

- 25 Preferably each wheel is mounted to a stub axle.

Preferably the lubrication delivery apparatus comprises a lubrication channel in each said stub axle.

Preferably an outer end of each said stub axle engages a longitudinally extending protection structure.

According to an third aspect of the present invention there is provided a sensor transportation apparatus to convey a sensor assembly through a wellbore, the apparatus comprising

a body to connect to said sensor assembly,

one or more wheels arranged to rotate on respective stub axles about an axis

5 substantially perpendicular to the longitudinal axis of said sensor assembly when the body is connected to said sensor assembly,

wherein one or more stub axles are supported by an orientation projection, or wherein one or more wheels are protected by an orientation projection.

10 Preferably, the orientation projection extends in front of the wheel which is connected to the respective stub axle.

Preferably the orientation projection extends longitudinally to a point behind the wheel which is connected to the respective stub axle.

Preferably the wheel extends above and below the orientation projection.

Preferably the orientation projection extends around the wheel and is connected to the body.

15 Preferably the orientation projection is removeably connected to the body.

According to a fourth aspect of the present invention there is provided a base for a guide device for guiding a tool-string comprising an elongate sensor assembly in a wellbore, the base comprising a first engagement portion for engaging an end of the tool-string and second
20 engagement portion for engaging a nose assembly, wherein the base is shaped such that a nose assembly engaged, in use, with the second engagement portion projects away from the base at an angle which is offset to a longitudinal axis of the tool-string engaged, in use, with the first engagement portion.

25 Preferably the base is formed as an integral element or component of one end of an orientation structure.

Preferably the base is hollow or has an aperture therethrough.

30 According to a fifth aspect of the present invention there is provided a guide device for guiding a tool-string in a wellbore, the tool-string provided, in use, with an orientation structure for orientating the sensor assembly in the wellbore, the guide device comprising a nose assembly having a base adapted to engage an end of the tool-string and a nose section

arranged to project away from the base at a fixed angle which is offset to a longitudinal axis of the tool-string when the nose assembly is engaged with said tool-string.

Preferably the fixed angle can be pre-set.

Preferably the fixed angle can be pre-set to one of a plurality of predefined settings.

- 5 Preferably the guide device comprises a locking pin engagable with the base and the nose assembly to lock the nose section at a selected angular setting.

Preferably the nose assembly comprises a locking member provided with a plurality of locking pin apertures and the base comprises a plurality of locking pin apertures.

Preferably the angle can be pre-set to between 1° to 60° .

- 10 Preferably the angle can be pre-set to between 1° to 45° .

Preferably the angle can be pre-set to between 1° to 20° .

Preferably the angle can be pre-set to between 1° to 9° , more preferably between 3° to 9° .

Preferably the nose section is formed from a resilient flexible material

Preferably the material is a resilient elastomer material.

- 15 Preferably the nose section is made from a material which is readily drillable by standard wellbore drilling equipment, for example nylon.

- 20 According to a sixth aspect of the invention there is provided a guide device for guiding a tool-string in a wellbore, the guide device comprising a nose assembly having a base adapted to engage an end of the tool-string and a nose section having a tip which is offset from a longitudinal axis of the tool string when the nose assembly is engaged with said tool-string.

- 25 According to a seventh aspect of the present invention there is provided a tool-string provided with the sensor transportation apparatus of the first, second or third aspect and/or the guide device of the fifth or sixth aspect.

According to an eighth aspect of the present invention there is provided a tool-string provided with a sensor transport apparatus for transporting an elongate sensor assembly through a wellbore, and an orientation structure, wherein

the sensor transportation apparatus comprises at least one engagement structure to connect the sensor transportation apparatus to the sensor assembly, and one or more wheels arranged to rotate about an axis of rotation substantially perpendicular to a longitudinal axis of the sensor assembly when the transportation apparatus is connected to the sensor assembly, and wherein

the orientation structure defines a form that has a transverse outline which has a rotational centre which is offset from a centroidal axis of the elongate sensor assembly.

Preferably, the engagement between the sensor transportation apparatus and the sensor assembly substantially prevents relative rotation between the sensor transportation apparatus and the sensor assembly about the longitudinal axis of the sensor assembly.

Preferably the orientation structure comprises a plurality of separate components.

Preferably the sensor transportation assembly comprises at least one of the separate components of the orientation structure.

According to a ninth aspect of the present invention there is provided a sensor transportation apparatus to convey an elongate sensor assembly through a wellbore, the sensor transportation apparatus comprising:

at least one engagement structure to connect the sensor transportation apparatus to the sensor assembly, and
at least one friction reduction element for reducing friction between the apparatus and the wellbore wall, and
an orientation structure defining a form having a rotational centre, wherein the rotational centre is substantially parallel to and offset from a centroidal axis of the elongate sensor assembly.

Preferably, the friction reduction element comprises at least one skid.

Preferably, the friction reduction element comprises at least one wheel arranged to rotate about an axis of rotation which is substantially perpendicular to a longitudinal axis of the sensor assembly when the transportation apparatus is connected to the sensor assembly.

According to a tenth aspect of the present invention there is provided a transportation apparatus comprising at least one engagement structure to connect the sensor transportation apparatus to the sensor assembly, and

one or more wheels arranged to rotate about an axis of rotation substantially perpendicular to a longitudinal axis of the sensor assembly when the transportation apparatus is connected to the sensor assembly, and

at least one protection structure which extends longitudinally around the wheel, to thereby substantially prevent a side of the wheel from contacting the wellbore wall, wherein the wheel extends above and below the protection structure.

Preferably each wheel is mounted on an axle, more preferably a stub axle.

Preferably each stub axle is engaged with a protection structure. Alternatively, the stub axle is integral with the protection structure.

Preferably the wheel is mounted on an axle, and the protection structure has a height which is at least equal to a diameter of the axle, but less than a radius of the wheel.

Preferably there is a clearance space between a surface of wheel which is at the radial extremity of the wheel, relative to the axis of rotation, and a body of the transportation apparatus, of at least 4mm.

Preferably there is a clearance space between a surface of wheel which is at the radial extremity of the wheel, relative to the axis of rotation, and an inner surface of the protection structure, of at least 4mm, or preferably at least 10mm, more preferably substantially 19mm.

Preferably the protection structure is an orientation projection.

According to an eleventh aspect of the present invention there is provided an apparatus for supporting, orienting and transporting an elongate sensor assembly in a wellbore, the apparatus comprising a body having at least one engagement structure releasably attachable to the sensor assembly and an orientation structure having one or more radial projections formed from the body corresponding to a substantially constant radius from a rotational centre,

wherein the rotational centre is offset from a centroidal axis of the elongate sensor assembly such that the eccentric mass of the sensor body causes the tool assembly to orientate toward the lower aspect of the wellbore.

- 5 According to a further aspect of the present invention there is provided the use of an apparatus of the first, ninth or eleventh aspects to maintain a logging tool in a preferred orientation within a wellbore.

- 10 According to a further aspect of the present invention there is provided a method of taking a wellbore measurement from an upper surface of a wellbore comprising positioning a sensor provided with an orientation structure as herein described inside the wellbore and operating the sensor to take the measurement.

- 15 According to a further aspect of the present invention there is provided an elongate sensor transportation apparatus substantially as herein described with reference to any one or more of the embodiments described herein and illustrated in the drawings.

- 20 According to a still further aspect of the present invention there is provided a guide device for guiding a tool-string substantially as herein described with reference to any one or more of the embodiments described herein and illustrated in the drawings.

- 25 The invention may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features, and where specific integers are mentioned herein which have known equivalents in the art to which the invention relates, such known equivalents are deemed to be incorporated herein as if individually set forth.

- 30 Further aspects of the invention, which should be considered in all its novel aspects, will become apparent from the following description given by way of example of possible embodiments of the invention.

Brief description of the drawings

An example embodiment of the invention is now discussed with reference to the drawings in which:

Figure 1 shows a isometric view of a sensor transportation apparatus as configured in accordance with a preferred embodiment of the invention.

Figure 2 shows a side view of the transportation apparatus of figure 1

Figure 3 shows a top view of the transportation apparatus of figures 1 and 2.

- 5 Figure 4 shows a cross-section along plane A-A of the transportation apparatus of figures 1-3, and shows the integral lubrication delivery apparatus. The wellbore wall is shown in outline.

Figure 5 shows a cross-section of an alternative embodiment along plane A-A. The wellbore wall is shown in outline.

- 10 Figure 6a shows a front view of a transportation apparatus provided in accordance with an alternative embodiment to that shown in figures 1 – 5, with the wellbore shown outline.

Figure 6b shows a perspective view of the transportation apparatus of Figure 6a.

- 15 Figure 7 shows a perspective view of a wireline tool-string engaged with both the transportation apparatus illustrated in respect to figure 6a and 6b in addition to the apparatus shown with respect to figures 1-4.

Figure 8 shows an isometric view of a further embodiment of the transportation apparatus.

Figure 9 shows a cross-section view of the embodiment shown in Figure 8.

Figure 10 shows a side view of a guide device provided in accordance with one embodiment of the invention.

- 20 Figure 11 shows a front perspective view of the guide device of Figure 10.

Figure 12 shows a rear perspective view of the guide device of Figure 10.

Figure 13 shows an exploded side view of another embodiment of the guide device.

Figure 14 shows an exploded perspective view of the guide device of Figure 13.

Figure 15 shows an isometric view of a base for a guide device.

Figure 16 shows a perspective view of an embodiment of an adjustable guide device set to a 3° angle.

Figure 17 shows the adjustable guide device of Figure 16 set to an 8° angle.

Figure 18 shows the adjustable guide device of Figure 16 set to a 15° angle.

5 Figure 19 shows exploded perspective view of another embodiment of an adjustable guide device.

Figure 20 shows an isometric view of an orientation structure which is provided with low friction skids.

10 Figure 21 shows an isometric view of another embodiment of an orientation structure which is provided with low friction skids.

Figure 22 shows an isometric view of another embodiment of an orientation structure which is provided with low friction skids.

Figure 23 shows an isometric view of another embodiment of an orientation structure which is provided with low friction skids.

15 Figure 24 shows a transverse cross-section of another embodiment of an orientation structure which is provided with low friction skids.

Figure 25 shows a transverse cross-section of another embodiment of an orientation structure which is provided with low friction skids.

20 Figure 26 shows a transverse cross-section of another embodiment of an orientation structure which is provided with low friction skids.

Figure 27 shows a transverse cross-section of another embodiment of an orientation structure which is provided with low friction skids.

Figure 28 shows a perspective view of a guide device and orientation structure mounted to a wireline logging tool-string.

25 Figure 29 shows a perspective view of the guide device of Figure 28 in combination with an alternative orientation structure which is provided with low friction skids, mounted to a wireline logging tool-string.

Figure 30 a perspective view of the guide device of Figure 28 in combination with two orientation structures which are provided with low friction skids mounted to a wireline logging tool-string.

5 Figure 31 a perspective view of the guide device of Figure 28 in combination with transportation apparatus mounted to a wireline logging tool-string.

Figure 32 shows a perspective view of a weight bar attached to a guide device according to Figure 19 and three transport apparatus according to Figure 41.

Figure 33 shows a perspective view of a guide device according to Figure 19 attached to a wireline tool-string, with a bowspring orientation structure.

10 Figure 34 shows a perspective view of another embodiment of the guide device.

Figure 35 shows a perspective view of a lubrication delivery apparatus.

Figure 36 shows a side cross-section view of the lubrication delivery apparatus of Figure 35 prior to charging with lubricant.

15 Figure 37 shows a side cross-section view of the lubrication delivery apparatus of Figure 35 after charging with lubricant

Figure 38 illustrates the pressures experienced by the lubrication delivery apparatus of figures 35, 36 and 37 when used in a particular instance, and

20 Figure 39 shows a transverse cross-section of a transportation apparatus with an integral lubrication delivery apparatus, with one ball bearing assembly mounted wheel shown and the opposite wheel and bearing removed.

Figure 40 shows a perspective view of the transportation apparatus of Figure 39 with the opposite wheel and bearing shown in partially exploded form.

Figure 41 shows a perspective view of the transportation apparatus in another embodiment.

25 Figure 42 shows a transverse cross-section of the transportation apparatus of Figure 41 provided with a modified lubrication delivery system, plain bearings and separate stub axles.

Figure 43 shows an isometric view of a transport apparatus configured as a tractor unit, with the jockey wheel in a raised position.

Figure 44 shows a front view of the tractor unit of Figure 43 in use in a wellbore, with a lower portion of the jockey wheel shown in hidden detail.

Figure 45 shows a front view of the tractor unit of Figure 43 in use in a wellbore with the jockey wheel in a lowered position, and with a lower portion of the jockey wheel shown in hidden detail.

Figure 46 shows a perspective view of another embodiment of an orientation structure.

Figure 47 shows a transverse cross-section of the orientation structure of Figure 46, shown in use in a wellbore (wellbore shown in hidden detail).

Figure 48 shows a perspective view of a guide device and orientation structure mounted to a wireline logging tool-string which has a sampling tool on its upper surface.

Figure 48 shows the same wireline logging tool-string as Figure 48, but with wheeled transportation providing the orientation.

Best modes for carrying out the Invention

Figures 1 - 4 show a number of views of the transportation apparatus 1 as provided. In this embodiment the transportation apparatus is arranged to connect over or on to the exterior surface of an elongate sensor assembly— shown in cross-section in Figure 4 as a tool-string 2 used in a wireline logging applications. Figure 4 also illustrates the position of the longitudinal axis 2a of the tool-string 2. As used herein the phrase “elongate sensor assembly” includes both the sensors themselves, and any equipment attached to the sensors such as weights, spacers, flexible couplings, positioning devices and the like.

Figure 4 illustrates the perimeter walls of a wellbore 3 which the tool-string 2 is to be conveyed down by the transportation apparatus 1. For the sake of convenience the wellbore 3 is shown with a substantially horizontal orientation, with its low side 3a shown at the bottom of the page.

The transportation apparatus 1 includes the main body 4, used to mount and locate the remaining elements or components of the apparatus.

Figures 1 through 3 show details of a pair of engagement structures formed in this embodiment by integral locking collars 5. These locking collars 5 are located at either end of the apparatus 1. Each collar 5 is arranged to partially enclose the exterior side wall surface of the tool-string 2,

allowing the transportation apparatus to be slid on and over the tool-string at any desired position along the length of the tool-string. Each collar includes a series of threaded holes 5a arranged to receive a screw which engages with a recess or blind hole in the exterior surface of the tool-string. These threaded holes 5a and an associated set of screws are used to lock the apparatus 1 to the tool-string 2 with a specific orientation. By arranging the collars 5 to only partially enclose the tool-string 2, the tool-string can be carried in a lower position than would be possible if the collars 5 were to completely encircle the circumference of the tool-string 2.

The main body of each of these collars also defines or forms a protection structure 14 which protects wheels deployed on either side of the body from impact and abrasion. These protection structures also serve to orientate the carriage and tool-string assembly in the wellbore. These protection structures can also prevent the space between the wheel and tool-string 2 catching on wellbore projections, such as the casing shoe, as the tool-string 2 is pulled up and out of the wellbore. As can be seen from at least Figure 1 these protection structures may incorporate an angled surface projection 13 which can act to position the sensor assembly and transportation apparatus in the wellbore and guide the tool down the wellbore.

Figures 1 through 4 also show the provision of a pair of wheels 6 which form part of the transportation apparatus 1. These wheels 6 are deployed laterally adjacent to one another on opposite sides of a tool-string. In the embodiment shown each wheel rotates on a stub axle 7. The wheels 6 have an axis of rotation which is perpendicular to a longitudinal axis of the elongate sensor assembly. In the embodiment shown, the wheels have a common axis of rotation.

As can be seen from these figures each wheel 6 has a diameter substantially greater than one of the diameter, width or height of the tool-string 2. This allows each of the wheels 6 to make radial contact with the wall of the wellbore 3 (that is, the radial edge of the wheels makes contact with the wellbore wall). The large relative size of each wheel 6 lifts the tool-string 2 from the low side surface of the wellbore 3, allowing the transportation apparatus 1 to convey the tool-string 2 down the wellbore 3. These large wheels allow the transportation apparatus 1 to roll along rugose borehole walls, up and over obstructing ledges which would normally impede the progress of the wireline logging tool-string 2. The wheels are widely spaced in the wellbore with a broad wheel track which allows the wheels to bridge over obstructing wellbore cuttings which tend to sit on the low side of the hole 3a. In preferred embodiments the wheels 6 extend below the main body 4 and below the sensor engaged with the apparatus, and thereby create a

minimum clear space of at least 10mm, more preferably at least $\frac{1}{2}$ inch, between the tool string 2 (and the body) and the bottom of the wellbore.

Figure 4 shows a cross-section view of the transportation apparatus of Figures 1 – 3 along the plane A-A shown in Figure 2 and 3. Figure 5 shows similar view to Figure 4 of a cross-section view of the transportation apparatus provided in a further embodiment. In the embodiment shown with respect to Figure 4 a double set of ball bearing assemblies 10 are used, whereas in Figure 5 the wheels rotate over a set of bushes 110.

Figures 4 and 5 show the positioning and arrangements of the axis of rotation of each wheel 6 as provided by stub axles 7. As can also be seen from these figures each of these stub axles 7 is located above the longitudinal axis 2a of the tool-string 2. The longitudinal axis is essentially the axial or longitudinal centre of mass of the tool-string. As the tool-string 2 is much heavier than the transportation apparatus 1 the tool-string 2 will tend to rotate into the orientation shown in Figure 4 where the transportation apparatus sits squarely on both wheels.

Identified on these figures is the rotational centre 15 of the transportation apparatus and the centroid 2a or centre of mass of the tool-string. In the embodiment shown each section of the tool-string has a centre of mass 2a approximately in the middle of the circular section of the tool-string. The rotational centre 15 of the transportation apparatus is located above the centroidal axis 2a of the tool-string. The centroidal axis of the tool-string, as that phrase is used herein, is formed by the line joining the centre of mass of each cross-section along the length of the tool-string. The axis of rotation of the wheels 6 preferably passes through the rotational centre 15.

The body 4 also mounts an orientation structure comprising at least one orientation projection 8. As can be seen from Figure 4 the orientation projection 8 assists in providing the tool-string 2 with a predictable orientation within the wellbore 3. If the tool-string 2 were to rotate to place the orientation projection 8 in contact with the low side of the wellbore 3 this would make the position of the combined transportation apparatus and connected tool-string 2 unstable. The orientation projection 8 therefore assists in positioning and orientating the tool-string in the arrangement shown in respect of Figure 4. The radially extreme edges of the wheels 6 and the protection structure 14 also form part of the orientation structure and assist in orientating the tool-string in the orientation shown in respect to Figure 4.

Figure 4 and 5 also show the form provided through a cross-section of the transportation apparatus made through all of its orientation projections 8, protection structures 14 and wheels

6. This form has a transverse outline (that is an outline or silhouette when viewed along the longitudinal axis of the sensor assembly) which has a rotational centre 15. The position of this rotational centre is the point which experiences the minimum vertical displacement when the form is rolled over a flat horizontal surface. In preferred embodiments the extremities of the orientation structure lie on a substantially circular imaginary curve centred on the rotational centre 15. The rotational centre of the transportation apparatus is offset from the centroid or centre of mass of the tool-string 2a.

The offset between the rotational centre of the transportation device and centre of mass of the tool-string ensures the assembly is orientated in the most stable position with the tool-string centre of mass below the rotational centre of the transportation apparatus. In this stable position the tool-string descends down the wellbore carried on the wheels of the transportation apparatus.

Referring in particular to Figure 4, in this embodiment the rotation of each wheel 6 is lubricated by a pressurised lubrication system 9 as it rotates about a double set of ball bearing assemblies 10. Figure 5 shows an alternative arrangement of the transportation apparatus where the lubrication system 9 acts on a set of bushes 110. This lubrication system 9 includes a lubricant reservoir 11 and pressurising plunger linked to the bearings 10 or bushes 110 by a pair of channels 12. In other embodiments the plunger may be replaced by a set of elastomer bellows, or by an elastic diaphragm, as is described further below. The plunger is maintained at a higher pressure than the surrounding wellbore fluids thereby force feeding lubricant to the wheel bearing or bush surfaces via the channels 12. Pressurisation of lubricant at this point in the apparatus prevents the entry of exterior contaminants from the wellbore 3.

Figures 6a and 6b show cross-section and isometric views of a transportation apparatus 201 provided in accordance with alternative embodiment to that shown in Figures 1 – 5. Figure 7 shows an isometric view of a wireline tool-string 2 engaged with both the transportation apparatus illustrated in respect to Figure 6a and 6b in addition to the apparatus shown with respect to Figures 1-4.

In the embodiment of the apparatus shown in Figure 6b the transportation apparatus 201 is arranged to engage with the bottom end or nose of a tool-string, thereby forming the leading component deployed down a wellbore. In this embodiment of the transportation apparatus 201 includes a single wheel 206 deployed as the leading component of the apparatus. This single

wheel can act as a hole finding assembly which can encounter and then roll over the top of obstacles immediately in the path of the tool-string. As can also be seen from Figures 6a and 6b the wheel 206 has a scalloped or concave profile, allowing it to bridge cuttings and obstacles encountered in the middle of the low side of the wellbore. The wheel is scalloped to ensure that only the edges of the wheel contact the wellbore wall resulting in a wide wheel track width, lower rolling friction and greater stability in the operating position.

The axis of rotation of the wheel 206 is above the centreline or longitudinal mid-line axis 202a of the tool-string which defines the longitudinal or axial centre of mass of the tool-string.

Furthermore, the apparatus 201 has an orientation structure which in this embodiment comprises multiple orientation projections 208 arranged to assist in maintaining this desired orientation of the tool-string. The orientation projections 208a also function as protection structures and protect the wheels. As can be seen in Figures 6a and 6b, one or more of the orientation projections 208a may also provide a mounting point for mounting the wheel 206 or wheel axle.

Figure 6a also shows the form provided through a cross-section view of the transportation apparatus 201 made through all of its orientation projections 208 and the end view of wheel 206. This form has a transverse outline with a rotational centre 207. The rotational centre is substantially parallel to and offset from a central axis of the elongate sensor assembly. The rotational centre of the transportation apparatus is offset from the centroid or centre of mass of the sensor assembly/tool-string 202a.

As with the embodiments described above, the offset between the rotational centre of the transportation device and centre of mass of the tool-string ensures the assembly is orientated in the most stable position with the tool-string centre of mass below the rotational centre of the transportation device. In this stable position the tool-string descends down the wellbore carried on the wheels of the transportation device.

Figure 7 shows how these two different implementations of the transportation apparatus can be deployed in conjunction with a single tool-string 2 to form a system for transporting the tool-string. The single wheeled apparatus 201 is used to lift the bottom end or nose of the tool-string while the paired wheels of the transportation apparatus 1 is used to carry the upper section of the tool-string 2. In another embodiment the transportation apparatus may be provided with a partial orientation structure, or no orientation structure at all. In these embodiments a separate component may carry some or all of the orientation structure. For example, a tool-string may

be provided with a transportation apparatus and one or more of the orientation structures shown in Figures 20, 21 or 33. The relative positioning of the separate components of the orientation structure along the sensor assembly is not critical, provided the overall form has a transverse outline with a rotational centre.

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Figures 8 and 9 show an alternative embodiment of the transportation apparatus 1a. This embodiment is configured for in-line connection to a sensor assembly, rather than mounting over the sensor or other sensor assembly component. As can be seen, the transportation apparatus 1a is provided with an orientation structure which includes a protection structure which includes removable orientation projections 8a which extend from in front of the wheels, around an exterior surface 6a of the wheels 6. These orientation projections 8a are shaped to assist in preventing material from clogging the space between the inside surface of the wheel 6b and the corresponding exterior surface of the main body 4a.

10

15 The width of the wheels 6 is preferably allows a clearance space between the inner surface of the wheel and the body, and between the outer surface of the wheel and the protection structure. In a preferred embodiment there is at least 4mm clearance between the operating surface of the wheels and the surrounding body and structure, as is described further below.

20 As is best seen in Figure 9, in this embodiment the orientation projections 8a also operate to support the outer ends of the stub axles 7 on which the wheels 6 are mounted. Those skilled in the art will appreciate that in an alternative embodiment (not shown) the stub axles 7 may be permanently connected to or integral with the removable orientation projections 8a, and may be engaged with and supported by the body 4a at their inner ends.

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As shown in the cross-section view of Figure 9, the wheel 6 diameter is greater on the inner face 6b than the outer face 6a to fit to the circular wellbore profile. This variation in wheel diameter causes some skidding at the contact surface between the wheel and the wellbore wall as the wheel rotates. Preferably the wheel should be narrow in order to minimise skidding contact and provide the least friction between the wheel and the wellbore wall as the wheel rotates down the wellbore. As can be seen from these figures, the width of the wheels 6 is less than about 20mm, preferably less than 18mm. In one embodiment shown in figure 9 the wheel width is 10mm. In another preferred embodiment shown in fig 42 the wheel width is 15mm.

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Figures 10 to 12 show a number of views of a guide device 21. The guide device 21 includes a nose assembly 22 formed from a nose section 23 and a base 27, as shown in Figures 10 and 11. Directly attached to the nose assembly 22 is an orientation structure 13, provided in the embodiment shown by an orientated stand-off 24.

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The orientated stand-off 24 provides a sleeve like structure with a hollow interior cavity 29 arranged to receive the leading end of the wireline logging tool-string. The orientated stand-off 24 includes a pair of friction reducing elements, in this case travel skids 26, which are oriented to slide along the downside wall of the wellbore through the action of a set of three orientation
 10 projections 8 provided by the orientated stand-off 24. In the embodiment shown the travel skids 26 are formed from a durable ceramic material which has low friction characteristics and resists abrasion, and the effects of high temperatures and high alkaline conditions found in a wellbore, for example alumina ceramic or polymetallic oxide thermosetting ceramic. In other embodiments suitable plastic or elastomer materials can also be used for this component. When provided
 15 with such friction reducing elements the orientated standoff 24 can function as a transportation apparatus, although wheels may be necessary for high deviation angles.

The travel skids, when provided, may be designed for use in a specific wellbore diameter, and may be shaped to conform to the curvature of the wellbore wall. The leading and trailing edges
 20 of the travel skids are contoured to form a gentle entry to allow the skids to “ski” over mudcake and cuttings debris.

The travel skids are positioned on the orientated standoff so that they do not travel over the lowermost section of the wellbore where cuttings are likely to accumulate. Rather, the skids
 25 straddle the cuttings and are positioned laterally in on the borehole wall, preferably between 30° and 45° from a vertical plane through the longitudinal axis of the sensor assembly.

As is described above with reference to Figures 1 to 4, the orientation projections 8 act to destabilise the position of the tool-string if placed in contact with the low side wall of a deviated
 30 wellbore. The substantial weight of the tool-string will ensure that if one of these projections is in contact with the down side wall, the tool-string will rotate back about it's longitudinal axis to an orientation placing the travel skids in contact with the downside wall.

As can be seen from Figures 10 to 12, the nose section 23 has a fixed offset angle relative to
 35 the longitudinal axis of any tool-string engaged with the interior cavity 29 of the orientated stand-

off 24. This arrangement provides the nose section 23 with a permanent upward inclination as the combined guide device and tool-string is deployed down a deviated wellbore. The upward inclination of the nose section ensures that the nose section tip 36 is elevated above the low side of the wellbore by a height 37, and is offset from and above the longitudinal axis of the tool-string. Hence obstacles on the low side wall of the wellbore are avoided by the guide device with the tool-string following immediately after the nose assembly.

Figures 13 and 14 show an exploded view of a further preferred embodiment of the guide device. The guide device includes a nose section 23, a base 27 and an orientation structure 24. All three components may be partially or fully integrated (not shown) or provided as discrete components as in the embodiment shown. In the embodiment shown the base 27 has a fixed angle relative to the tool-string longitudinal axis and is attached directly to the leading end of the wireline logging tool-string. The nose section 23 is attached to the terminal end of the base 27. The orientation structure slides over and is attached to the logging tool-string such that the orientation projections are aligned to orient the nose section 23 upwards as the tool-string descends down the wellbore.

The nose section 23 can be implemented in different lengths, thereby altering the height 37 from the bottom of the wellbore to the nose section tip 36. The offset of the nose section tip 36 is pre-set depending on the wellbore size and conditions.

An enlarged view of one embodiment of the base 27 is shown in Figure 15. The base 27 may be used with some nose sections of the prior art in order to provide improved performance, although those skilled in the art will appreciate that the tool-string will require some type of orientation structure, for example an orientated stand-off, bow spring excentraliser and/or transportation apparatus as described herein, in order to orientate the nose away from the base of the wellbore.

The base 27 has a first engagement portion 25a for engaging an end of a tool-string, and a second engagement portion 25b for engaging a nose assembly. The engagement portions have centrelines which are angularly offset, such that a nose assembly engaged, in use, with the second engagement portion 25b has a centreline which extends at an angle to the centreline of a tool-string engaged with the first engagement portion 25a, as can be seen in Figure 13. The offset angle maybe any suitable angle, depending on the length of the nose assembly and the diameter of the wellbore, as is described further below.

In this embodiment the first engagement portion 25a is provided as a collar which fits over a standard boss provided at the end of a tool – string, and the second engagement portion 26b is provided as a hollow boss having standard outer dimensions for engaging a collar provided at the end of a prior art nose assembly. The base 27 is itself hollow, or at least has an aperture therethrough, to allow wiring to be passed through the base 27 if necessary.

Figures 16 to 18 show different configurations of angle nose assembly capable of being used in the embodiments shown in Figures 28-33. The nose assembly can rotate or pivot about an axle 38 which is engaged to the base 27, so that the angle of the nose assembly can be pre-set to a selected angle before the tool-string is lowered into the wellbore. The base 27 has a locking arm (not shown) extending inside the nose section 23. The nose section is set at a fixed pre-set angle from the longitudinal tool axis by a locking pin extending through one of three locking points 30 to connect with the locking arm. The nose section pivoting connection 38, combined with the locking point 30, secures the nose section to the base 27 at a fixed angle from the tool-string longitudinal axis. The selection of locking point 30 will determine the offset angle and hence the heights 37 of the nose section tip 36 above the low side of the wellbore.

As can be seen from Figures 16-18, each fixed angle setting of the nose assembly 22 provides the nose section 23 with a different offset angle. The offset angle implemented by each nose section 23 increases from 3° in Figure 16, 8° in Figure 17 and 15° in Figure 18. Those skilled in the art will appreciate that the pre-set angle may be adjusted depending on the diameter of the wellbore, the required standoff of the tool-string in the wellbore, and the length of the nose assembly, to locate the end of the nose assembly in a suitable position in the wellbore. Longer nose assemblies will require shallower angles, and shorter nose assemblies will require larger angles, for a given wellbore diameter and standoff distance. The angle is selected such that the tip of the nose section is above the longitudinal axis of the tool-string.

Figure 19 shows an alternative embodiment of an adjustable nose assembly. In this embodiment the nose section 23 is provided with a locking member 31. A pin or axle 38 extends through apertures in the base 27 and the locking member 31. The base 27 has a vertically orientated longitudinal slot 33 to accommodate the locking member 31.

The end of the locking member 31 is provided with a plurality of locking pin apertures 34a which are parallel to the aperture for the axle 38. Locking pin apertures 34b are also provided in the

base 27. The locking pin apertures 34a, 34b are arranged such that one of the apertures 34a in the locking member 31 is aligned with one of the apertures 34b in the base 27 when the nose section 23 is set to one of a predetermined number of angles. The nose section 23 is held at the required angle by inserting a locking pin 34 through apertures 34a, 34b. In a preferred embodiment the apertures 34a, 34b are arranged to allow the nose section to be set to angles of 3 degrees, 6 degrees or 9 degrees to the centreline of the tool-string, although other embodiments may provide angles up to 60°, 45° or 20°. Angles as low as 1° may be used in some embodiments, although often this will require the use of an inconveniently long nose assembly.

Figures 20-23 show perspective views of two different types of orientation structure provided in accordance with a yet further embodiment. These figures again illustrate the different forms of orientated stand-off 24 which are to be provided as orientation structures of a guide device.

These different orientated stand-offs cater for different sizes or diameters of wellbore size. These figures also illustrate different arrangements of orientation projections 8.

The stand-offs shown in Figure 20 and Figure 22 employ a set of four orientation projections 8, while the stand-offs 24 shown in Figures 21 and 23 employ three orientation projections only.

The orientation projections 8 have tapered extremities 35 to assist in guiding the tool-string around or past obstacles encountered in a wellbore.

The oriented standoffs have a hole 29 passing through the body of the device. This hole is offset from the nominal centre of the standoff. The oriented standoff is designed to slip over a cylindrical wireline logging tool which is then fixed within the hole 29, and below the rotational centre of the stand-offs. The tool-string will therefore be conveyed with a predictable orientation, which in turn will impart an upward orientation or angle to the nose section of the nose assembly.

Figures 24-27 show a series of transverse or cross-section views of a number of different implementations of orientation structures. These orientation structures are capable of being used in the embodiments shown with respect to Figures 10-19 and 28-30. Identified on these figures is the rotational centre 15 of the orientation structure and the centroid 2a or centre of mass of the tool-string. In the embodiment shown each section of the tool-string has a centre of mass 2a approximately in the middle of the circular section of the tool-string.

These figures also show the form provided through a transverse cross-section of each orientated stand-off made through all of its orientation projections 8 (which in these embodiments also corresponds to the transverse outline). The form provided by this cross section therefore has rotational centre 15. The rotational centre of the oriented standoff is offset from the centroid or centre of mass of the tool-string 2a.

In a deviated wellbore the tool-string will naturally rotate about its longitudinal axis to seek the most stable position. The most stable position is where the tool-string and oriented standoff assembly has the lowest centre of gravity. The lowest centre of gravity is where the tool centroid is below the centre of rotation. In this stable position, the guide device nose section is orientated upwards, the low-friction travel skids of the orientated standoff are in contact with the low side of the wellbore and the logging tool sensors can be optimally orientated to measure the preferred side of the wellbore.

These views show various arrangements of orientated stand-offs with different numbers of orientation projections 8. As can be seen from these figures an orientated stand-off may be implemented with a number of different arrangements of orientation projection. These projections can also be formed with a variety of lengths to accommodate a range of different wellbore diameters.

As can also be seen from the embodiment displayed in these figures, in these embodiments the extremities of each of the orientation projections 8 define sections of the perimeter of a substantially circular curve centred on a point above the centre of gravity of any tool-string engaged with the orientation structure. In the embodiments shown, each of the projections 8 is arrayed at an equal angle to that of its neighbouring projections. However, the travel skids 26 are preferably arranged to have a greater spacing between them than the spacing between the other orientation projections, in order to maximise the stability of the apparatus when it is in its preferred orientation.

Figures 28 - 31 demonstrate the use of the guide device. The figures show the guide device displayed in Figures 16-18 and an associated wireline logging tool-string. These figures show a variety of embodiments where an orientation structure 24 or structures are connected to a tool-string 2 at positions remote from a nose assembly 22.

Figure 28 illustrates the use of a single orientation structure, shown in this embodiment by an orientated stand-off 24 which is connected some distance away from a nose assembly 22.

Figure 29 illustrates a similar alternative arrangement of orientated stand-off 24 which is arranged to be deployed in a larger diameter wellbore than the stand-off shown with respect to Figure 28, and hence has the nose 23 set at a larger angle and uses a larger diameter orientation structure.

Figure 30 shows in a further embodiment which employs a pair of orientated stand-offs 24 located at two different positions along the length of a tool-string 2.

In the embodiment shown in Figure 31 a pair of transportation apparatus 1 are used to orientate the guide device 21. The transportation apparatus 1 again act to provide the tool-string 2 with a predictable orientation and to impart an upward inclination to the nose section 23 of the nose assembly 22, as is described above with reference to Figures 1-9.

Figure 32 shows a variation of the system shown in Figure 31. In this embodiment the transport apparatus 1 are attached to a weight bar 39. The weight bar 39 is attached to a tool-string 2, preferably by a flexible coupling. This allows the transport apparatus 1 and guide device 21 to be used with any type of tool-string, including those with sensors that must be touching the borehole wall, those with irregular profiles, and exceptionally large diameter tools relative to the wellbore diameter. Tools that require lateral movement in the wellbore during operation may also be used, for example sampling tools which move across wellbore when set, and imager tools that are run in hole collapsed and open out to be centralised when deployed.

Figure 33 shows another embodiment in which the orientation structure includes the use of a bowspring excentraliser as an orientation projection 8. Bowspring excentralisers are well known to those skilled in the art.

Referring next to Figure 34, an embodiment of the guide device 21 is shown which has a base 27b which has provision for mounting a temperature sensor 40 and/or other sensor means which require direct contact with the fluid in the wellbore. The base 27b has a guard structure 41 comprising two parallel ribs 42 arranged on either side of the sensor 40, the ribs 42 extending at least as far from the body of the base 27b as the sensor 40. This allows the temperature sensor probe to extend into the central area of the wellbore without risk of damage resulting from contact with the wellbore wall. In this position the temperature sensor assembly

is well flushed by the movement of the tool through the wellbore fluids, and is not liable to be so covered with mudcake as to unduly affect response times. To ensure a fast response to temperature changes, preferentially the temperature sensor probe housing is made from a high thermal conductivity, low corrosion material such as low Beryllium Copper alloy, silver or gold.

- 5 The temperature sensor probe is preferably thermally insulated from the base and body of the guide device 21.

By positioning the probe at the end of the nose of the tool-string the probe measures wellbore temperature before that temperature is influenced by the temperature of the tool-string itself, and
10 mixing of well fluids caused by the tool-string movement.

A pressure sensor may also be provided (additionally or alternatively to the temperature sensor). The provision of a pressure sensor which does not become clogged with mudcake or other debris allows the change of depth of the tool to be monitored. In this way the operator can
15 ensure that the tool-string is proceeding into the wellbore at a rate which is consistent with the rate at which the cable feed is operating. Feeding cable at too high a rate can result in the cable becoming tangled and the tool-string becoming difficult to remove from the wellbore.

The sensor 40 is positioned so as to be behind the upwardly angled nose section 23. Because
20 the orientation structures keep the nose section 23 angled upward, and the nose section 23 is held at a fixed angle when in use, the sensor 40 does not come into contact with the sides of the wellbore, and is shielded by the nose section 23 from any steps or shelves in the wellbore walls. In the embodiment shown the guide device 21 is provided with the pre-set angle adjustment system described above with reference to Figure 19, and shows the cover plate 43 which is
25 used to retain the locking pin 34.

Referring next to Figures 35-37, a lubrication delivery apparatus which is suitable for use with the transportation apparatus 1 is generally referenced by arrow 300.

30 The apparatus 300 is formed from a base structure, provided in this embodiment by a block element 52. An inlet channel 53 is formed within the interior of the block element, as is an outlet channel 55. The inlet channel 53 terminates in an inlet port, formed in this embodiment by a grease nipple or valve 54. The outlet channel 55 leads to an outlet port 56 which allows for the free flow of lubricant, typically grease, from the apparatus 300.

The apparatus 300 has an elastic diaphragm 37 which covers most of the upper flat surface of the base structure 52. The diaphragm 57 is connected to and sealed against the base by a substantially rigid protective housing 58. The housing 58 also includes a port 59 to allow ambient pressure to act on the exterior side of the elastic diaphragm. The port also provides an opening for inspection of the diaphragm 57.

As can be seen from Figure 37, a reservoir 60 is formed between the interior surface of the diaphragm 57 and the surface of the base which it covers. Figure 36 shows the arrangement of the apparatus prior to this reservoir being filled with lubricant.

To charge or fill the reservoir 60, a suitable flowable lubricant, typically grease, is forced into the reservoir 60 through the grease nipple or valve 54 via a grease gun, oil pump or similar. The injected lubricant stretches the elastic diaphragm 57. The tension induced in the elastic diaphragm 57 due to this stretching, results in the lubricant in the reservoir 60 being maintained at a pressure higher than the ambient pressure acting on the exterior side of the elastic diaphragm 57.

Figure 37 shows the apparatus after the reservoir 60 is charged or filled with lubricant injected through the grease nipple or valve 54 and inlet channel 53. The lubricant is pressurised in the reservoir by the stretched elastic diaphragm 57. In this embodiment the elastic diaphragm has been stretched to twice its original area (elongation 100%). The lubricant is delivered from the pressurised reservoir 60 to the mechanical system via outlet channel 55 and outlet port 56.

In the embodiment shown the diaphragm 57 is formed from silicone rubber, allowing it to be deformed and expanded to accommodate the pressurised lubricant pumped into the reservoir 60. The elastic nature of the diaphragm allows it to store elastic potential energy imparted by the pressurised lubricant supply to maintain the pressure of lubricant held within the reservoir. This stored energy is slowly released by the diaphragm pushing lubricant out through the outlet channel 55 and outlet port 56 to lubricate a further mechanical system (not shown) to be deployed and an underground wellbore. One example of a mechanical system which can be lubricated is a bearing assembly, as is described further below.

The tension in the elastic diaphragm 57 maintains the lubricant at an incremental pressure over and above the ambient pressure acting on the exterior surface of the elastic diaphragm 57. This over pressurisation of lubricant by the apparatus 300 provides a supply of lubricant to the

mechanical system being lubricated and also prevents the entry of contaminants from the wellbore into the mechanical system being lubricated.

In a variety of additional embodiments the flow of lubricant through the mechanical system may be controlled by a metering valve. The flow of lubricant may alternatively be limited by an oil or grease seal on the exit of the mechanical system. The mechanical system may also incorporate a bush bearing where flow of lubricant is limited by the close fit the bearing.

In the example shown in Figure 38 the ambient pressure (P) act on the exterior surface of the diaphragm 57 may be 14.7 psia. The additional pressure (δP) exerted on the lubricant by the stretched elastomer diaphragm 57 may be 3psia. Hence the lubricant reservoir pressure is 17.7 psia. Thus the pressure within the lubricated system is 3 psi higher than its surroundings.

An imperfect seal between the mechanical system which is supplied with lubricant by the apparatus 300 and its surroundings will result in lubricant leaking out to the ambient surroundings. The constant flow of lubricant out of the mechanical system prevents contaminants from entering the mechanical system. When the device is operating in a wellbore at 4000m below surface, the ambient pressure may be in the order of 6000 psia. The stretched elastomer diaphragm 57 will still exert an additional 3psi pressure on the lubricant and hence the lubricant pressure is 6003 psia. Pressure within the mechanical system is now 6003 psi, 3 psi above the surroundings, hence the mechanical system is protected from the entry of contaminants.

As the depth changes in the wellbore, so too does the ambient hydrostatic pressure. Unlike a spring/piston type system, the elastomer diaphragm has no sliding parts and is able to instantaneously adjust to small changes in ambient pressure, in this example, keeping lubricant pressure at 3psi above the surroundings. Because there are no sliding parts, and no friction inherent in the system, it will be effective at preventing contaminants entering the mechanical system even at very small differential pressure (δP) from the stretched elastomer diaphragm.

Figures 39 to 41 show a transportation apparatus which is provided with an alternative implementation of the lubrication delivery apparatus 300. The lubrication apparatus shown in the embodiment of Figures 39-41 is equivalent to that discussed above with respect Figures 35-38 other than for the inclusion of a pair of outlets channels and ports as discussed below, and

performs the function of the pressurised lubrication system 9 described above with reference to Figures 4 and 5.

In the embodiment shown with respect to Figures 39-41, the base structure of the lubrication delivery apparatus 300 is formed by a portion of the body of the transportation apparatus 1. As in the previously described embodiment, the base structure effectively forms a block like element which defines within its interior a pair of outlet channels 55 terminating in output ports 56. These outlet ports 56 are used to deliver lubricant to the bearings 61 of the transportation apparatus 1, as shown in Figures 39 and 40.

As best seen in Figure 39, each outlet port 56 feeds a chamber 62 formed between the end of the stub axle 7 and a cover 63 which is attached to the outer surface of the wheel 63. The chamber 62 extends to the open side of the bearing 61, thereby allowing pressurised lubricant to flow into the bearing 61.

A suitable lubricant seal 64 is provided on the inner side of the bearing 61. This arrangement allows the use of commercially available deep groove ball bearing assemblies 61 in the wellbore environment, the constant positive pressure on the inside of the oil/grease seal 64 ensuring that no water or other contaminants can penetrate the oil/grease seal 64.

The transportation apparatus shown in Figures 40 and 41 is provided with a protection structure in the form of removable orientation projections 8a which extend around an exterior surface 6a of the wheels 6, as described above with reference to Figures 8 and 9. These orientation projections 8a extend substantially longitudinally from in front of each wheel to behind the wheel, and are shaped to assist in preventing material from clogging the space between the inside surface of the wheel 6b and the exterior of the main body, and support the outer ends of the stub axles 7 on which the wheels 6 are mounted, as is described above. As can be seen in these figures, the wheels 6 extend above and below the longitudinally extending orientation projection 8a. The longitudinally extending orientation projection 8a is preferably relatively slender, for example having a height which is greater than the diameter of the axle on which the wheel is mounted, but less than the radius of the wheel. As is best seen in Figure 41, a clearance space is provided between the radially extreme edge or surface 67 of the wheel (relative to the axis of rotation) and the longitudinal orientation projection 8a, although the profile of the wheel may mean that the clearance is reduced towards the centre of the wheel.

This clearance, along with the slender aspect of the protection structure, assists in preventing the area inside the protection structure from becoming clogged with debris from the wellbore surface. In a preferred embodiment there is a clearance of at least 4mm between radially extreme surface 67 of the wheel (that is, the surface of the wheel which is normally in contact
 5 with the wellbore wall) and the exterior of the main body, and at least 4mm clearance between the radially extreme edge of the wheel and the interior of the protection structure. In the embodiment shown the interior of the protection structure has a clearance of at least 15mm, more preferably at least 19mm.

- 10 The longitudinally orientated orientation projection preferably has a central axis which is substantially coincident with the rotational axis of the wheel.

Figure 42 shows another embodiment of a transportation apparatus 1 with a modified lubrication delivery apparatus. In the embodiment shown the transportation apparatus 1 uses plain
 15 bearings with a flanged bush bearing member 61a. The lubrication apparatus 300 has two outlet channels and at least four outlet ports 56. No lubricant seal is required for this embodiment.

In this embodiment the stub axles 7 are formed as separate components to the body (in contrast
 20 to the embodiments shown in Figures 4 and 5 wherein the stub axles are integrally formed with the body). However, the channels 55 still run through each stub axle.

In a preferred embodiment the arrangement of the bearings is such that some pressurised lubricant is able to escape past the bearings. This constant, but controlled, escape of lubricant
 25 acts to transport debris and contaminants away from the bearings.

The stub axles 7 are supported by the longitudinally extending orientation projections 8a.

Figures 43 to 45 show another embodiment of the transportation apparatus, in this embodiment
 30 configured as a tractor unit.

The tractor unit 400 comprises a transportation apparatus with a plurality of main wheels 6, in this case four main wheels 6, and at least one jockey wheel 6c. The tractor unit preferably comprises an orientation structure having at least two orientation projections. Those skilled in
 35 the art will appreciate that the jockey wheel 6c also functions as an orientation projection.

The jockey wheel 6c is preferably mounted to an adjustable mounting means (not shown) for raising and lowering the jockey wheel relative to the main body 4a. One or more of the main wheels 6, and preferably each of the main wheels 6, are connectable to a drive means, typically one or more electric motors. In a preferred embodiment the main wheels 6 are provided with clutch means (not shown) which can disengage the main wheels 6 from the drive means to allow the main wheels 6 to freewheel when the borehole is sufficiently steep.

In use, one or more tractor units 400 are connected in line with the tool-string. The tool-string and tractor 400 can free-wheel down the wellbore until the tool-string can no longer descend under gravity alone. At this point the jockey wheel 6c is raised and extends upward to make contact with the top of the wellbore wall 3 (best seen in Figure 44) thereby increasing contact pressure on the driven wheels 6. The clutch is engaged and the main wheels 6 are driven by the drive means. In this way the tool-string can be transported along wellbores at very high deviation angles, up to and including substantially horizontal wellbores. The tractor unit is preferably provided with a guide device 21 as described above. One or more transportation means 1 may also be used as required to assist in transporting the tool-string.

Figures 46 and 47 show another embodiment of an orientation structure which is provided as an orientated standoff 24, similar to those shown in Figures 20 and 21. In the embodiment shown in Figures 46 and 47, the engagement between the standoff 24 and the tool-string is similar to that used with the transportation apparatus described above with respect to Figures 1-3. The engagement structure 5 is in the form of a locking collar. The collar 5 is arranged to partially enclose the exterior side wall surface of the tool-string 2, allowing the standoff to be slid on and over the tool-string at any desired position along the length of the tool-string. Each collar includes a series of threaded holes 5a arranged to receive a screw which engages with a recess or blind hole in the exterior surface of the tool-string. These threaded holes 5a and an associated set of screws are used to lock the apparatus 1 to the tool-string 2 with a specific orientation. By arranging the collars 5 to only partially enclose the tool-string 2, the tool-string can be carried in a lower position than would be possible if the collars 5 were to completely encircle the circumference of the tool-string 2, and allows an increased clear space 65 (preferably at least 10mm, more preferably at least ½ inch) between the lower orientation projections 8, which in this embodiment comprise low friction skids, This clear space allows the cutting debris which has collected on the bottom surface of the bore to pass under the tool-

string 2. As with the other orientation structures described above, the structures shown in Figures 46 and 47 have a centre of rotation which is offset from the centroid of the tool-string 2.

Figures 48 and 49 show a variant of the system shown in Figure 32. In this embodiment the tool-string is provided with a prior art sampling tool 66 for taking samples and pressure measurements. Because the tool-string 2 is connected to at least one orientation structure, whether an orientated standoff 24 or a transportation apparatus 1, the sampling tool can be orientated to take its sample from the high side of the wellbore. This has the advantage that the high side of the wellbore tends to be less damaged by the grinding action of the drillpipe as it rotates and reciprocates during the drilling operation. This damage can affect the porosity and permeability of the wellbore wall in particular, and can lead to unrepresentative measurements, or at least delays in obtaining representative measurements. By using the apparatus shown, this sampling process is improved.

Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”.

Where in the foregoing description, reference has been made to specific components or integers of the invention having known equivalents, then such equivalents are herein incorporated as if individually set forth.

Although this invention has been described by way of example and with reference to possible embodiments thereof, it is to be understood that modifications or improvements may be made thereto without departing from the spirit or scope of the appended claims.

Claims

1. A sensor transportation apparatus to convey an elongate sensor assembly through a wellbore, the sensor transportation apparatus comprising:
 - 5 at least one engagement structure to connect the sensor transportation apparatus to the elongate sensor assembly, and
 - one or more wheels arranged to rotate about an axis of rotation perpendicular to a longitudinal axis of the elongate sensor assembly when the sensor transportation apparatus is connected to the elongate sensor assembly, and
 - 10 an orientation structure defining a form having a transverse outline which has a rotational centre, wherein the rotational centre is offset from a centre of mass of the elongate sensor assembly so that, in use, the sensor transportation apparatus is oriented in a most stable position with the centre of mass of the elongate sensor assembly below the rotational centre.
- 15 2. The sensor transportation apparatus of claim 1 wherein the lateral extremities of the orientation structure lie on a circular imaginary curve which is centred at the rotational centre.
3. The sensor transportation apparatus of claim 1 wherein, in use, the sensor transportation apparatus has only one stable orientation.
- 20 4. The sensor transportation apparatus of claim 1 wherein the wheels are shaped and dimensioned such that, in use, contact between each wheel and a wall of the wellbore is at a surface of the wheel which is at the radial extremity of the wheel, relative to the axis of rotation.
5. The sensor transportation apparatus of claim 1, wherein each said wheel extends below
 - 25 the elongate sensor assembly and the engagement structure when the sensor transportation apparatus is in a horizontal orientation.
6. The sensor transportation apparatus of claim 5 wherein, in use the wheels carry the engagement structure and the elongate sensor assembly at least 10mm above a wall of the wellbore.
- 30 7. The sensor transportation apparatus of claim 1, wherein a rotational axis of each said wheel is above the centre of mass of the elongate sensor assembly when the sensor transportation apparatus is in a horizontal orientation.

8. The sensor transportation apparatus of claim 1, wherein the rotational axes of the wheels are parallel and coplanar.
9. The sensor transportation apparatus of claim 8 wherein at least two of the wheels have a common axis of rotation.
- 5 10. The sensor transportation apparatus of claim 1 wherein the wheels are mounted on stub axles.
11. The sensor transportation apparatus of claim 1, wherein the orientation structure comprises at least one laterally extending orientation projection.
- 10 12. The sensor transportation apparatus of claim 11, wherein the orientation structure further comprises at least one of the wheels.
13. The sensor transportation apparatus of claim 1, wherein the wheels run on bearings and the apparatus further comprises a lubrication delivery apparatus which provides a lubricant to the bearings at a pressure which is greater than ambient wellbore pressure.
- 15 14. The sensor transportation apparatus of claim 1, wherein the engagement structure engages an exterior surface of the elongate sensor assembly.
15. The sensor transportation apparatus of claim 14, wherein the engagement structure partially encloses the exterior surface of the elongate sensor assembly without completely encircling the elongate sensor assembly.
- 20 16. The sensor transportation apparatus of claim 1, wherein a portion of the elongate sensor assembly extends below the engagement structure when the sensor transportation apparatus is in a horizontal orientation.
17. The sensor transportation apparatus of claim 1 wherein the engagement structure is adapted for in-line connection to the elongate sensor assembly.
- 25 18. The sensor transportation apparatus of claim 17 further comprising a drive means, wherein at least one of the wheels is connectable to the drive means.
19. The sensor transportation apparatus of claim 18 comprising a clutch between the drive means and each driven wheel.
20. The sensor transportation apparatus of claim 17 comprising a jockey wheel mounted to adjustable mounting means for selectively raising or lowering the jockey wheel.

21. The sensor transportation apparatus of claim 20, wherein, when the adjustable mounting means are raised, the jockey wheel engages an opposite side of the wellbore to the wheels.
- 5 22. The sensor transportation apparatus of claim 1, wherein the elongate sensor assembly comprises a sampling tool oriented to take a sample from a high-side of the wellbore with the sensor transportation apparatus in the most stable position.
- 10 23. A sensor transportation apparatus to convey an elongate sensor assembly through a wellbore, the sensor transportation apparatus comprising at least one engagement structure to connect the sensor transportation apparatus to the elongate sensor assembly, and one or more wheels arranged to rotate on bearings about an axis of rotation perpendicular to a longitudinal axis of the elongate sensor assembly when the sensor transportation apparatus is connected to the elongate sensor assembly, and a lubrication delivery apparatus which provides a lubricant to at least one of the
15 bearings at a pressure which is maintained at greater than ambient wellbore pressure.
24. The sensor transportation apparatus of claim 23 wherein the bearing comprises a ball bearing assembly or a bush bearing assembly.
- 20 25. The sensor transportation apparatus of claim 23 wherein the lubrication delivery apparatus comprises a reservoir comprising an elastic diaphragm.
26. The sensor transportation apparatus of claim 25, wherein an exterior side of the elastic diaphragm is subject to ambient pressure.
27. The sensor transportation apparatus of claim 23 wherein each wheel is mounted to a stub axle.
- 25 28. The sensor transportation apparatus of claim 27 wherein the lubrication delivery apparatus comprises a lubrication channel in each said stub axle.
29. The sensor transportation apparatus of claim 27 wherein an outer end of each said stub axle engages a longitudinally extending protection structure.
- 30 30. A sensor transportation apparatus to convey an elongate sensor assembly through a wellbore, the sensor transportation apparatus comprising at least one engagement structure to connect the sensor transportation apparatus to the elongate sensor assembly, and one or more wheels arranged to rotate about an axis of rotation

perpendicular to a longitudinal axis of the elongate sensor assembly when the sensor transportation apparatus is connected to the elongate sensor assembly, and at least one protection structure which extends longitudinally around the wheel, to thereby prevent a side of the wheel from contacting a wall of the wellbore, wherein the wheel extends above and below the protection structure.

- 5
31. The transportation apparatus of claim 30 wherein each wheel is mounted on a stub axle.
32. The transportation apparatus of claim 31 wherein each stub axle is engaged with the protection structure, or is integral with the protection structure.
- 10 33. The transportation apparatus of claim 30 wherein the protection structure has a height which is less than a radius of the wheel.
34. The transportation apparatus of claim 30 wherein there is a clearance space between the surface of the wheel which is at the radial extremity of the wheel relative to the axis of rotation, and a body of the sensor transportation apparatus, of at least 4mm.
- 15 35. The transportation apparatus of claim 30 wherein there is a clearance space between the surface of the wheel which is at the radial extremity of the wheel relative to the axis of rotation, and an inner surface of the protection structure, of at least 4mm.
36. The transportation apparatus of claim 30 wherein the protection structure is an orientation projection.
- 20 37. A guide device for guiding an elongate sensor assembly in a wellbore and an orientation structure for orientating the guide device in the wellbore, the guide device comprising a base adapted to engage an end of the elongate sensor assembly and a nose section having a tip which is offset from a longitudinal axis of the elongate sensor assembly when the guide device is engaged with the elongate sensor assembly, and wherein the nose section with the tip projects from the base at a fixed angle to the longitudinal axis of the elongate sensor assembly when the guide device is engaged with the elongate sensor assembly.
- 25 38. The guide device of claim 37 wherein the fixed angle can be pre-set.
39. The guide device of claim 38 wherein the fixed angle can be pre-set to one of a plurality of predefined settings.
- 30 40. The guide device of claim 39 comprising a locking pin engageable with the base and the nose assembly to lock the nose section at a selected angular setting.

41. The guide device of claim 40 wherein the nose assembly comprises a locking member provided with a plurality of locking pin apertures and the base comprises a plurality of locking pin apertures.

42. A tool-string provided with:

- 5 a) the sensor transportation apparatus of any one of claims 1 to 36, or
- b) the sensor transportation apparatus of any one of claims 1 to 36 and the guide device of any one of claims 37 to 41.

43. A tool-string assembly comprising an elongate sensor assembly, a sensor transportation apparatus for transporting the elongate sensor assembly through a wellbore and an orientation structure, wherein

10 the sensor transportation apparatus comprises at least one engagement structure to connect the sensor transportation apparatus to the elongate sensor assembly, and one or more wheels arranged to rotate about an axis of rotation perpendicular to a longitudinal axis of the elongate sensor assembly when the sensor transportation

15 apparatus is connected to the elongate sensor assembly, and wherein

 the orientation structure defines a form that has a transverse outline which has a rotational centre which is offset from a centre of mass of the elongate sensor assembly so that in use the tool-string assembly is orientated in a most stable position with the centre of mass of the elongate sensor assembly below the rotational centre.

20 44. The tool-string assembly of claim 43 wherein the engagement structure prevents relative rotation between the sensor transportation apparatus and the elongate sensor assembly about the longitudinal axis of the elongate sensor assembly.

45. The tool-string assembly of claim 44 wherein the orientation structure comprises a plurality of separate components.

25 46. The tool-string assembly of claim 45 wherein the sensor transportation apparatus comprises at least one of the separate components of the orientation structure.

47. The tool-string assembly of claim 43, wherein the elongate sensor assembly comprises a sampling tool oriented to take a sample from a high-side of the wellbore with the sensor transportation apparatus in the most stable position.

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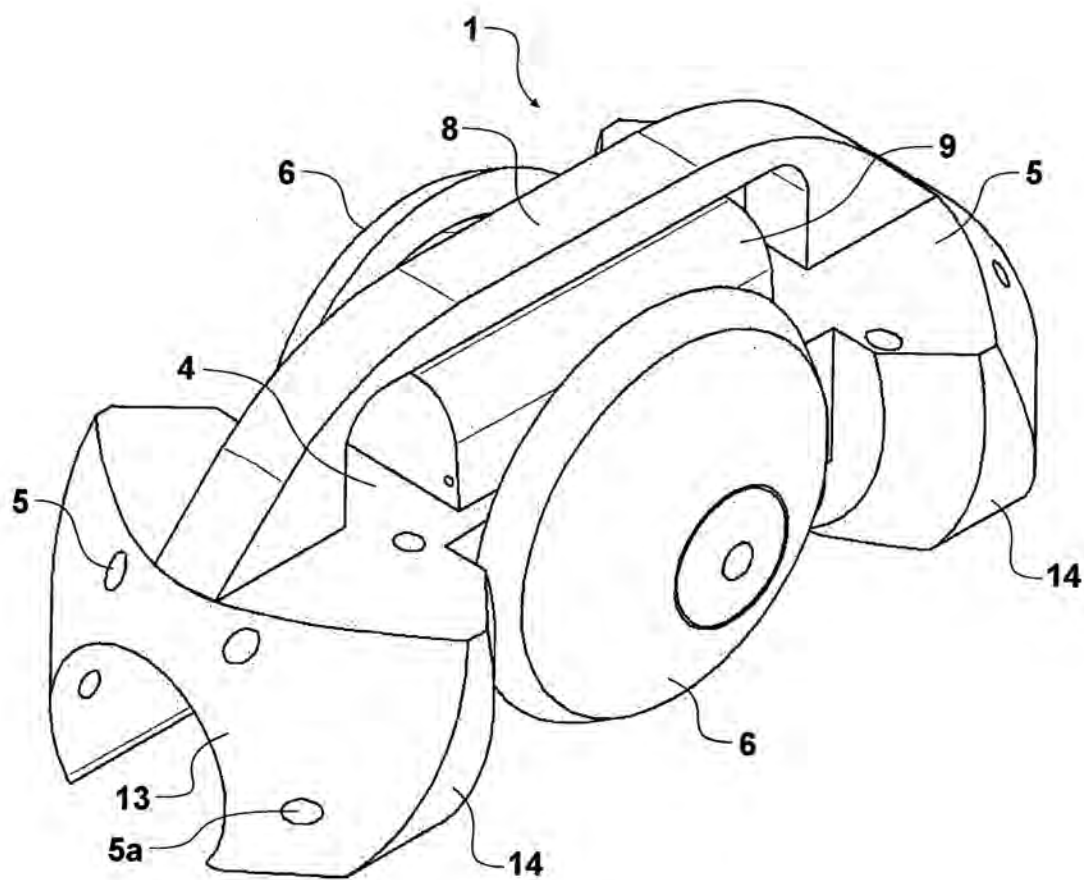
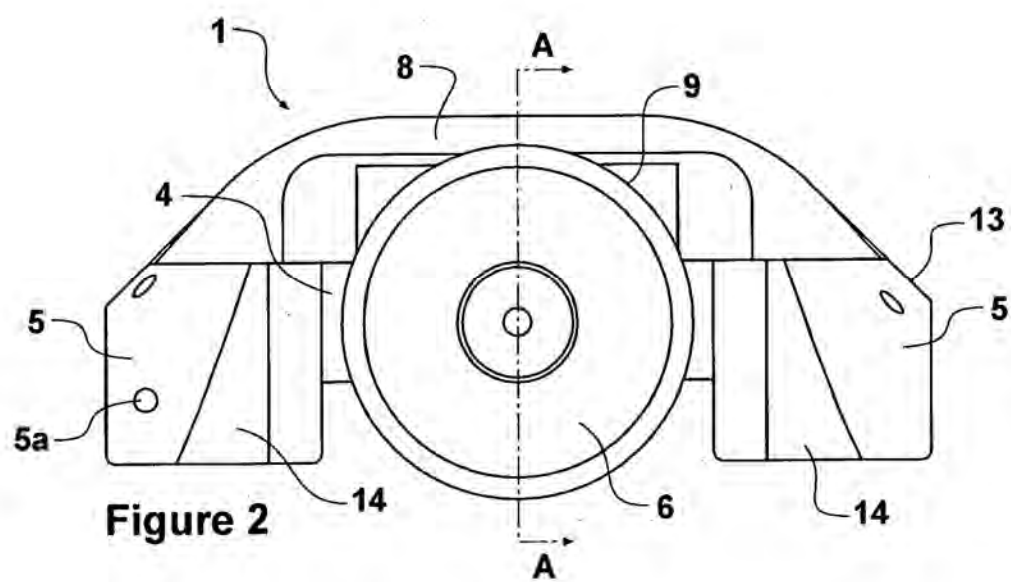


Figure 1

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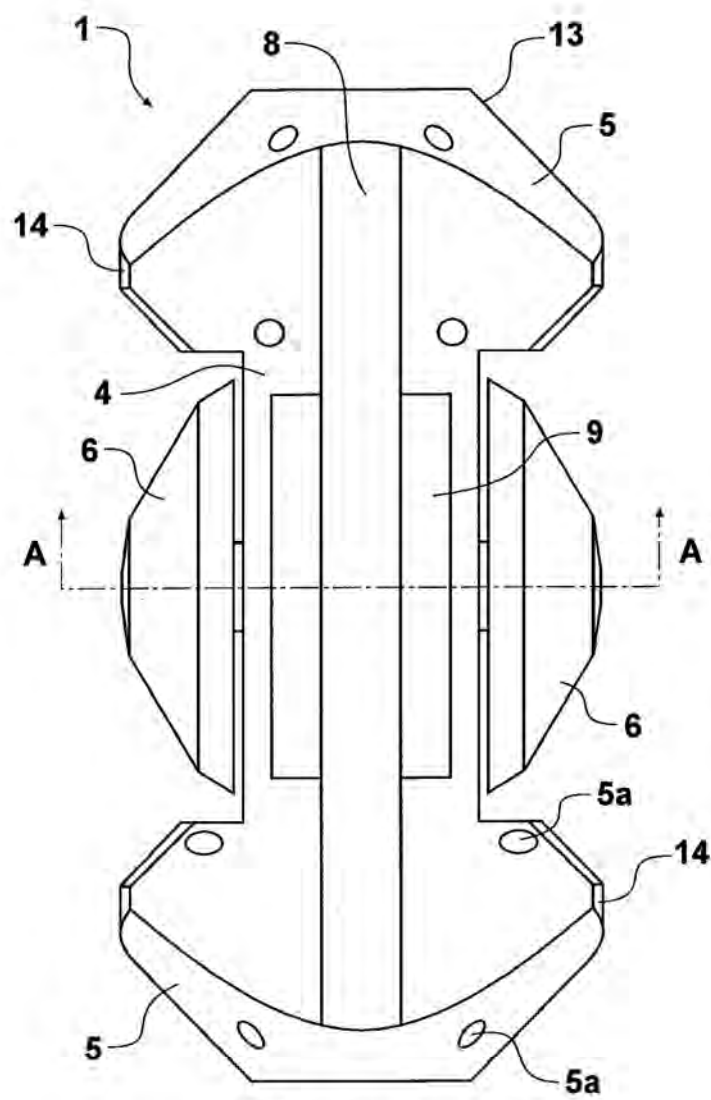
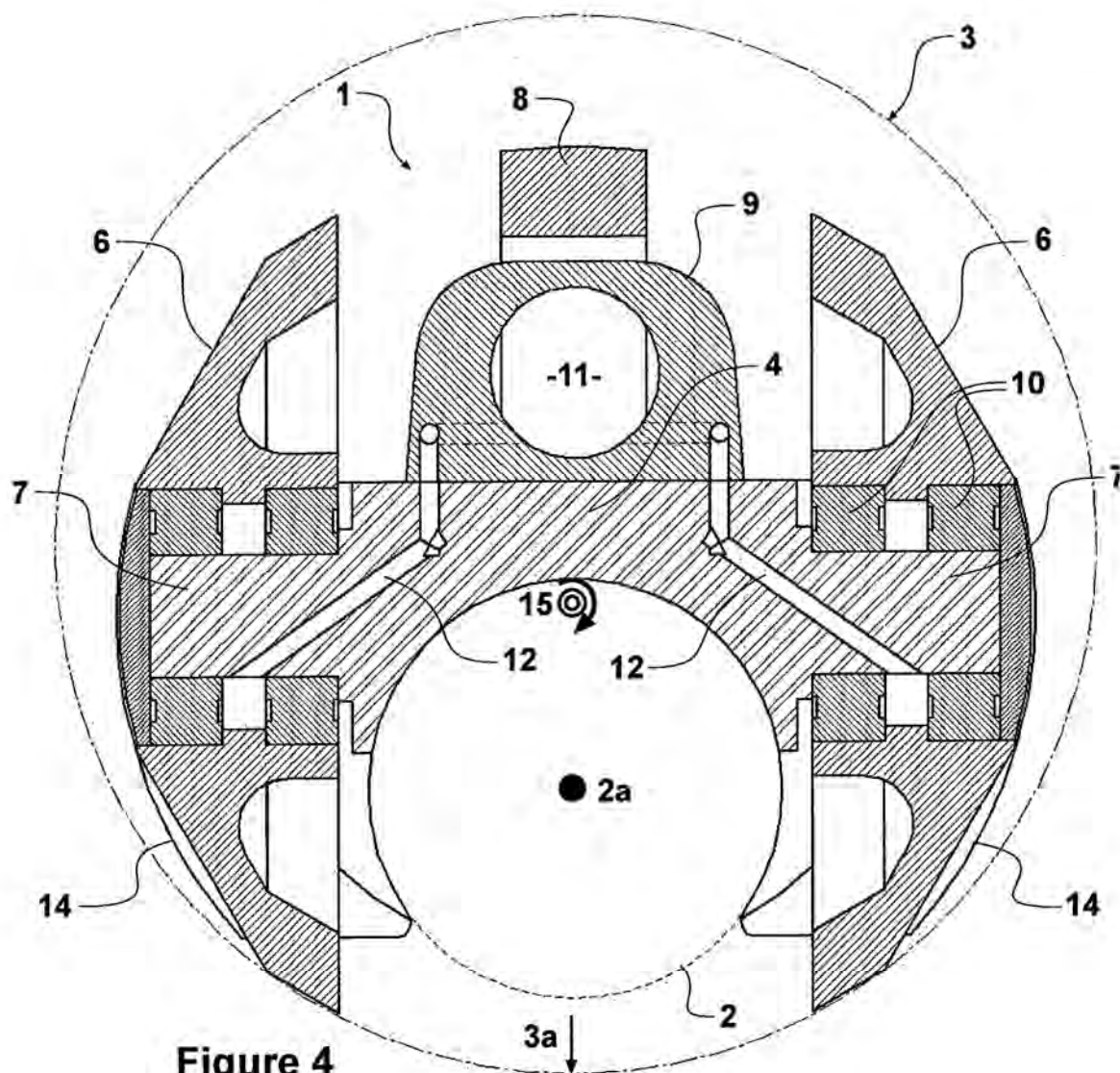
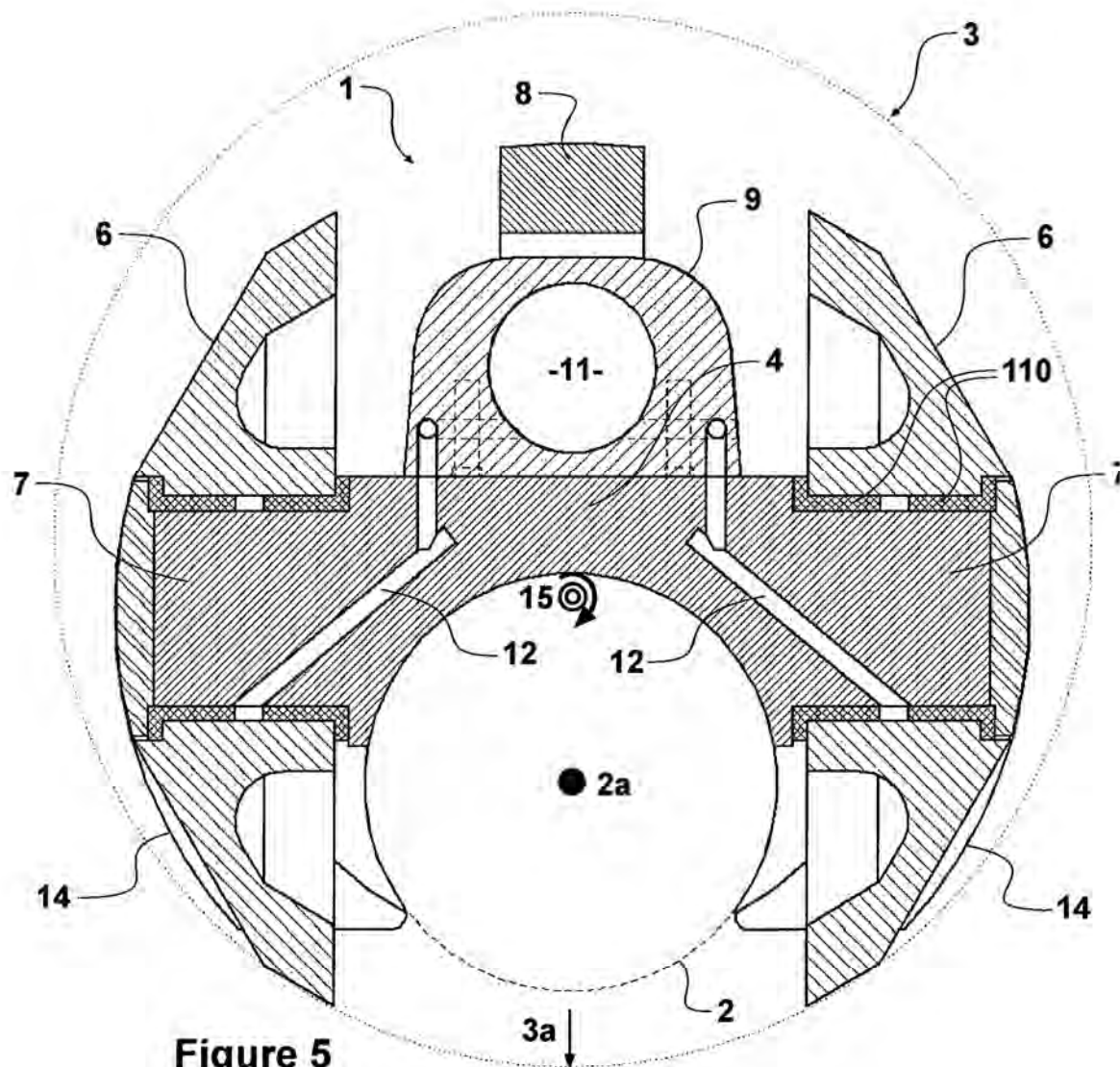


Figure 3

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Section A-A

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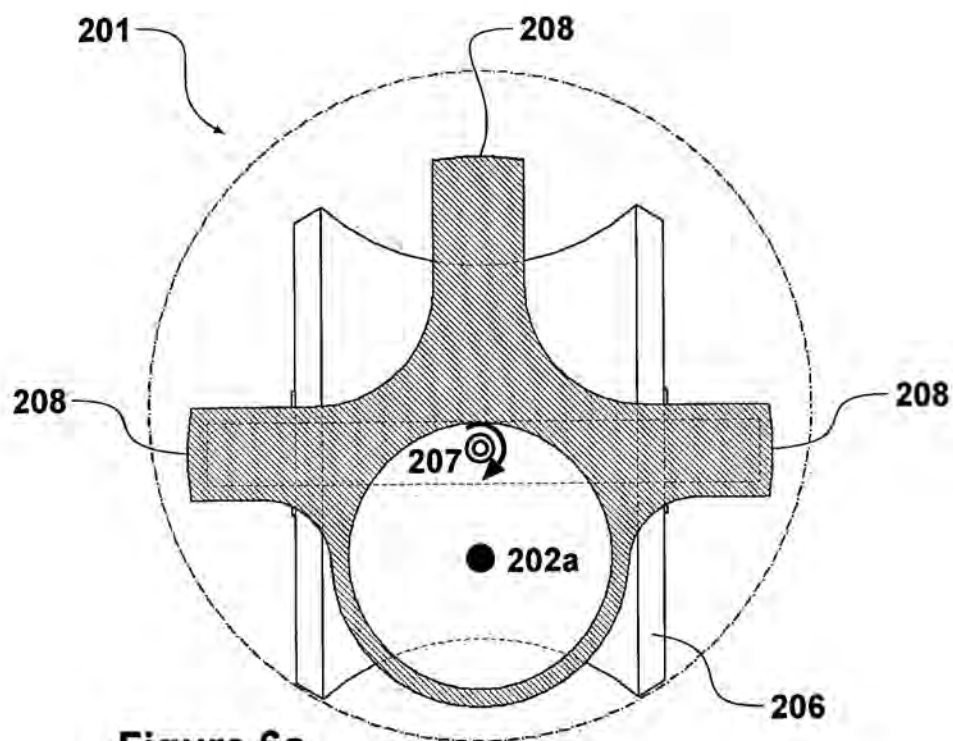


Figure 6a

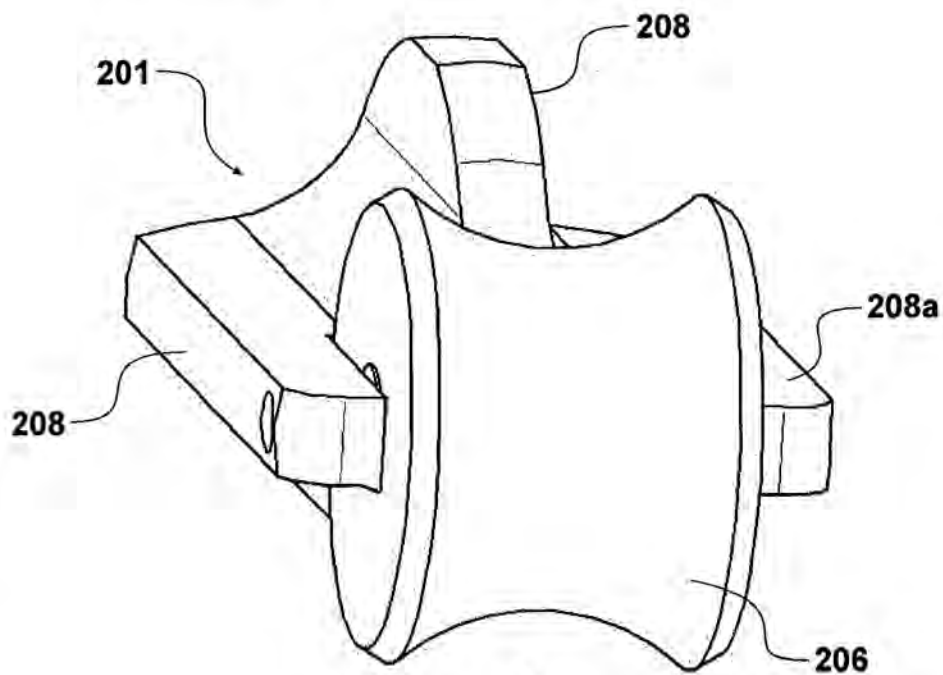


Figure 6b

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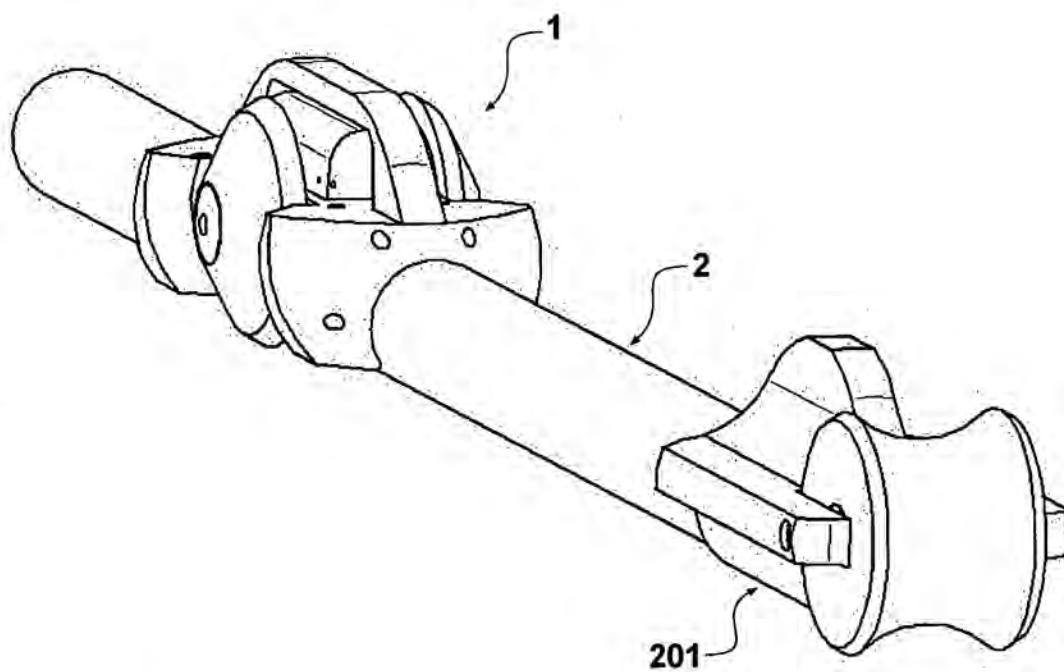


Figure 7

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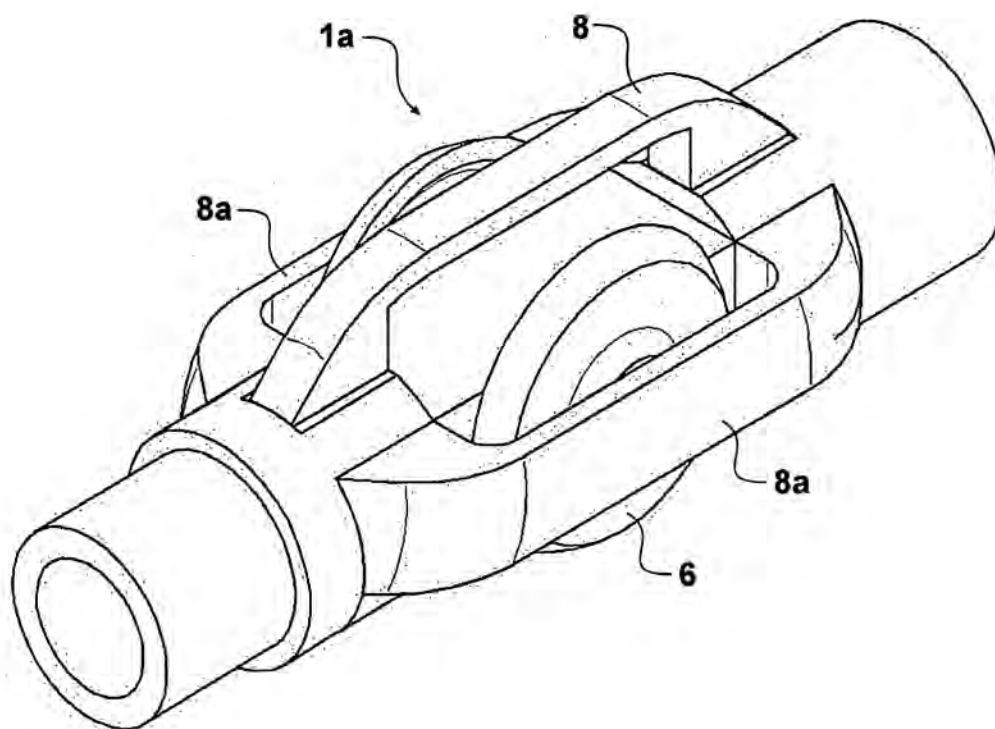


Figure 8

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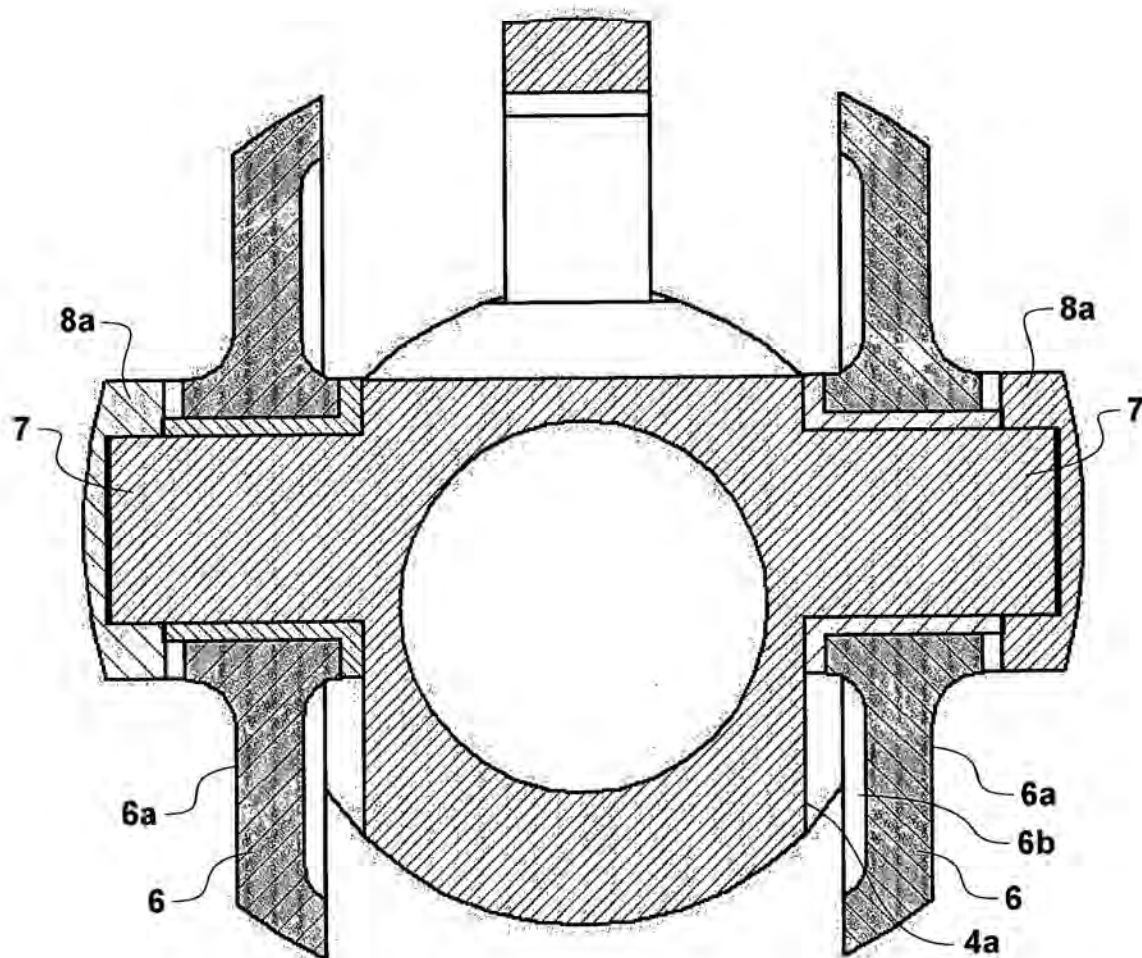
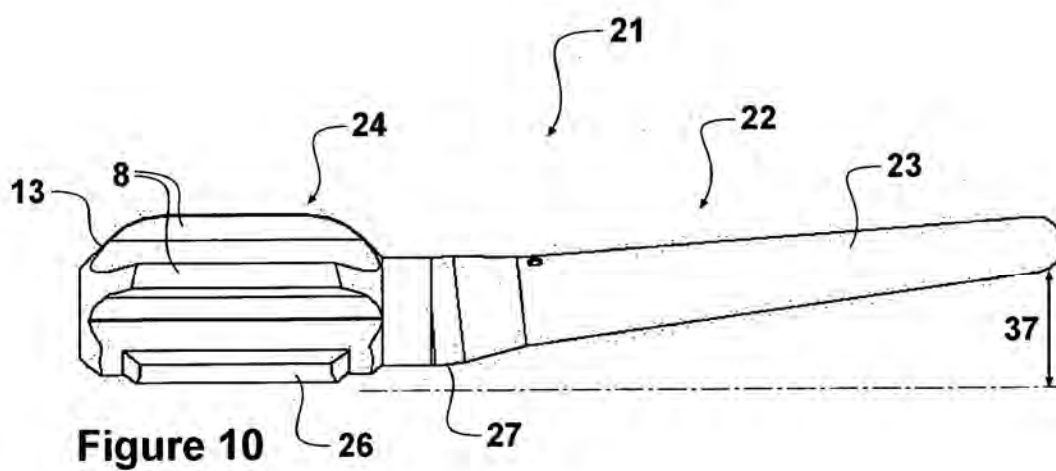


Figure 9

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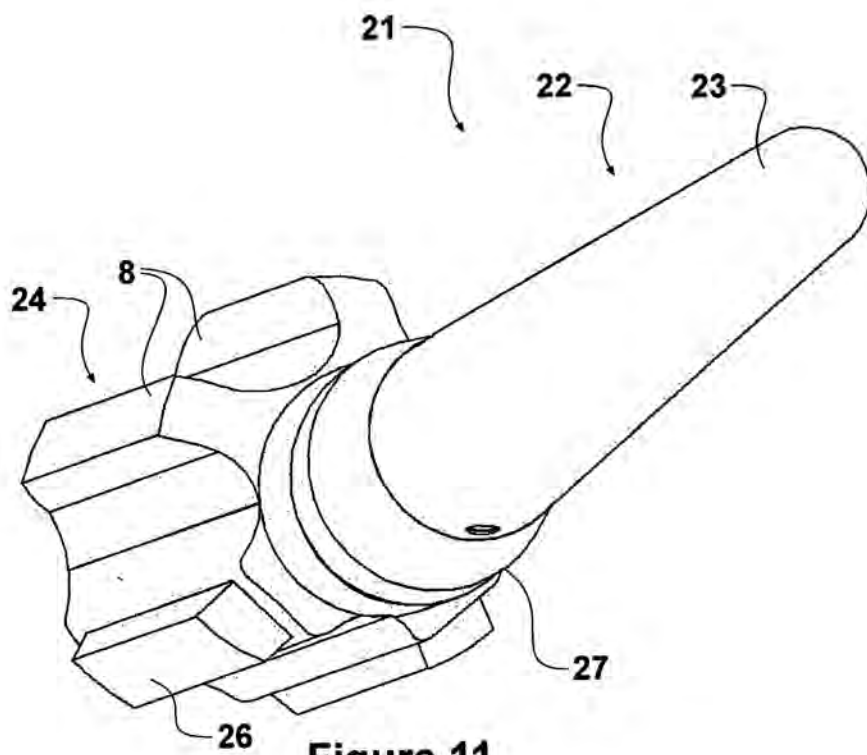


Figure 11

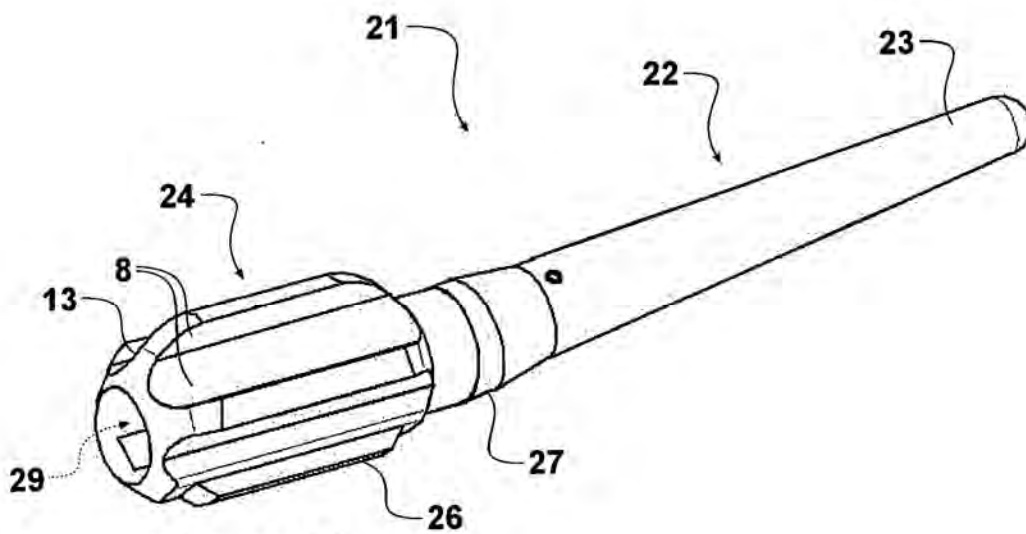


Figure 12

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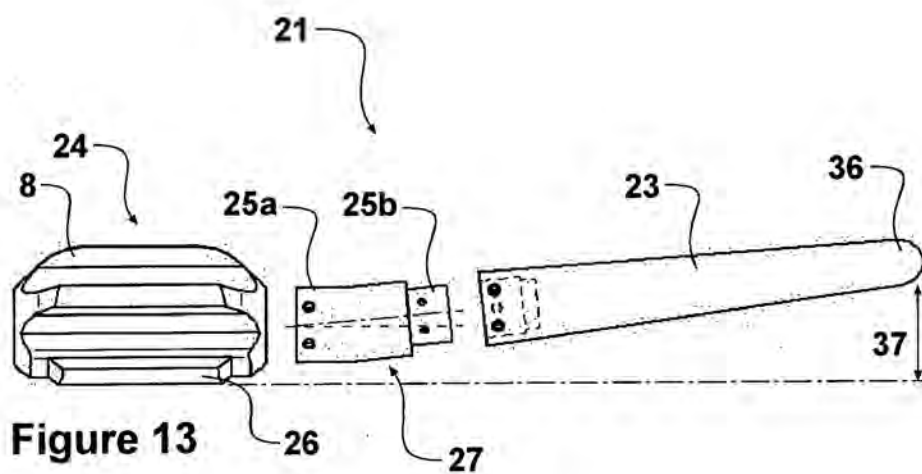


Figure 13

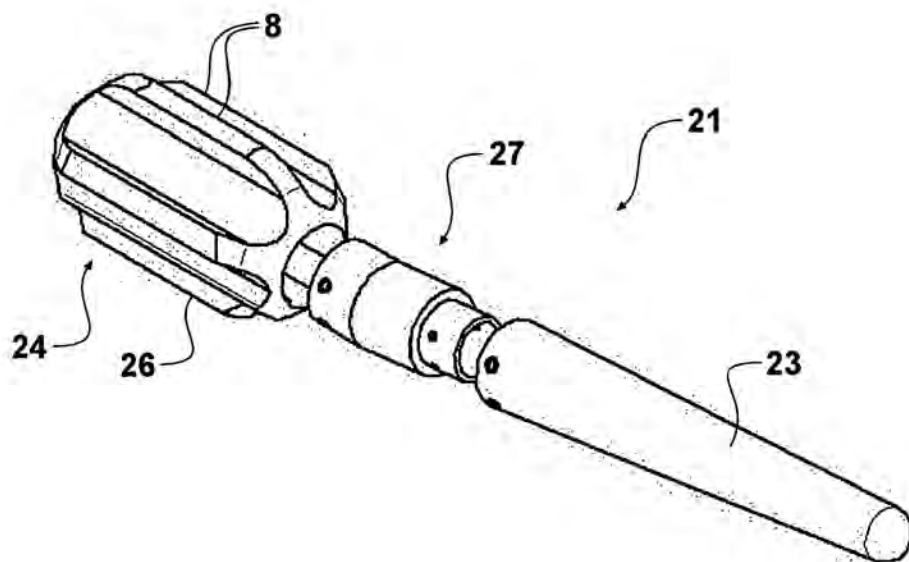


Figure 14

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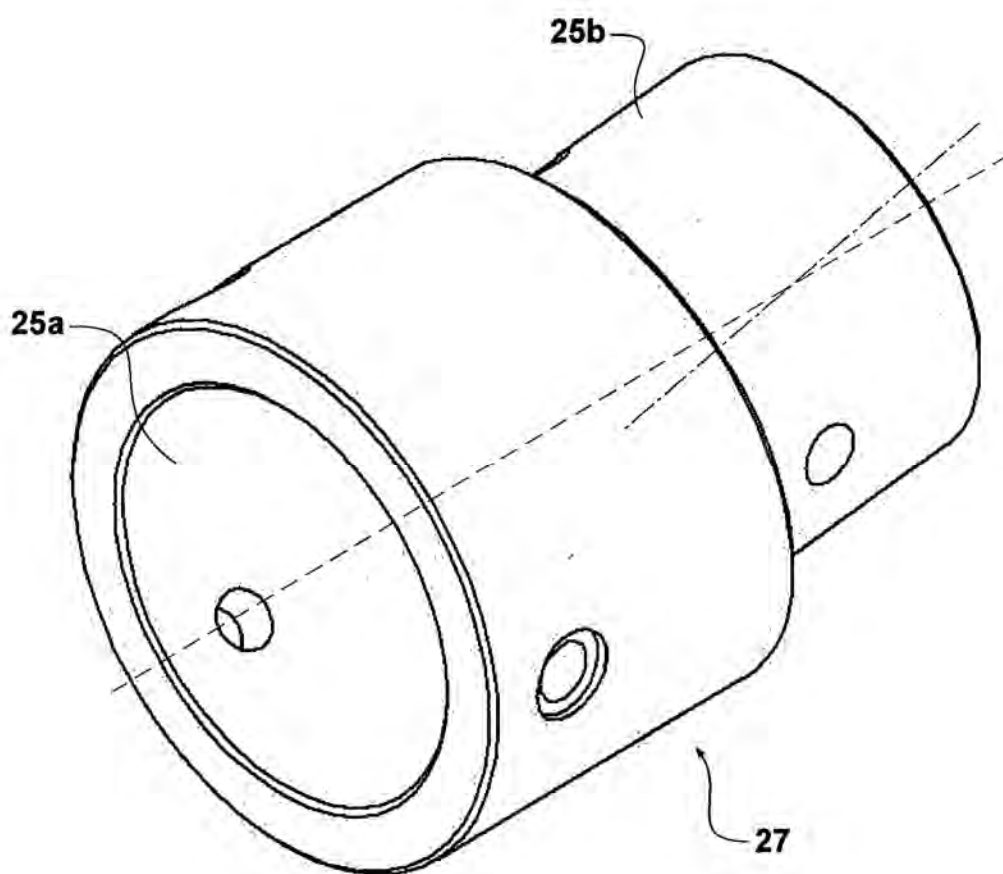


Figure 15

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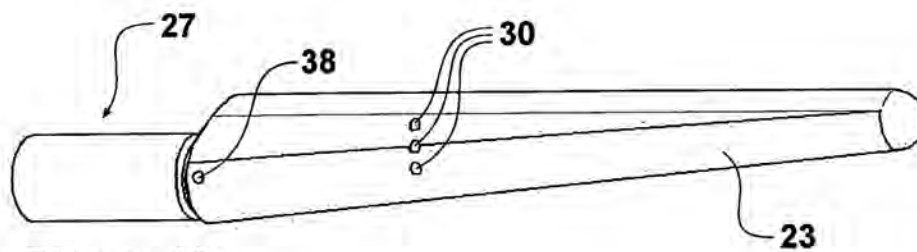


Figure 16

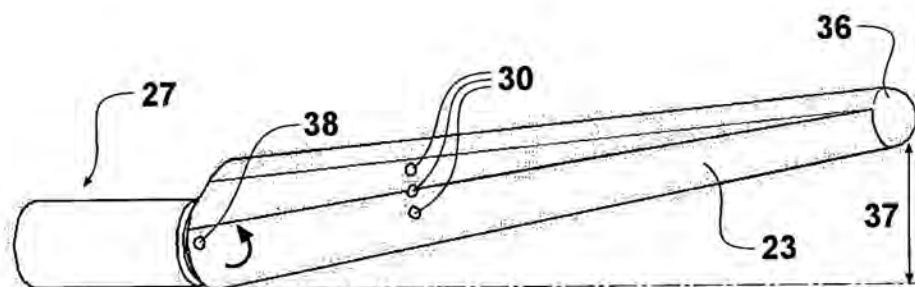


Figure 17

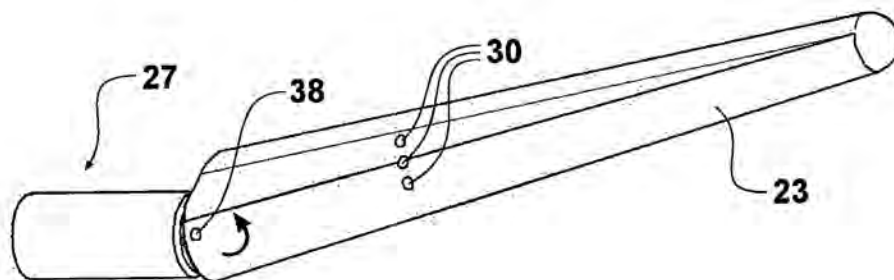


Figure 18

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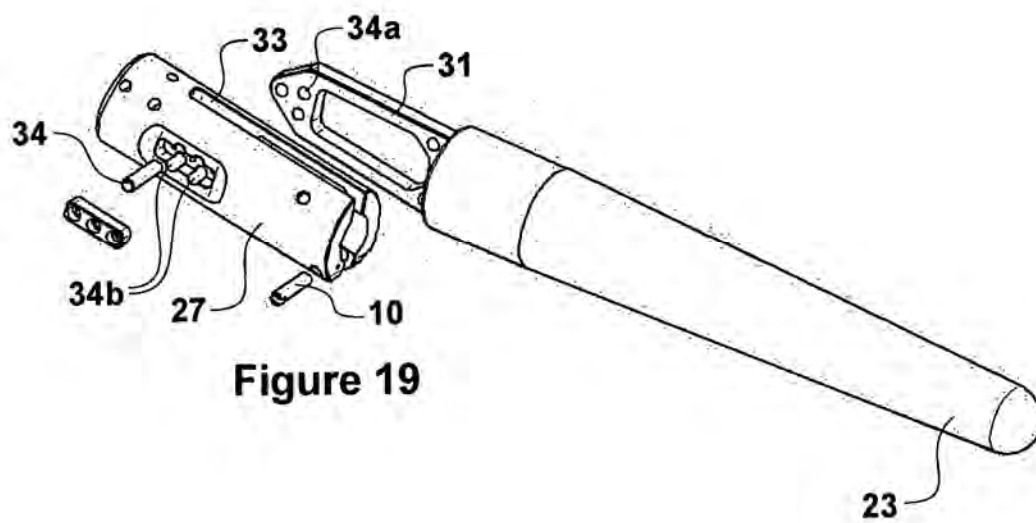


Figure 19

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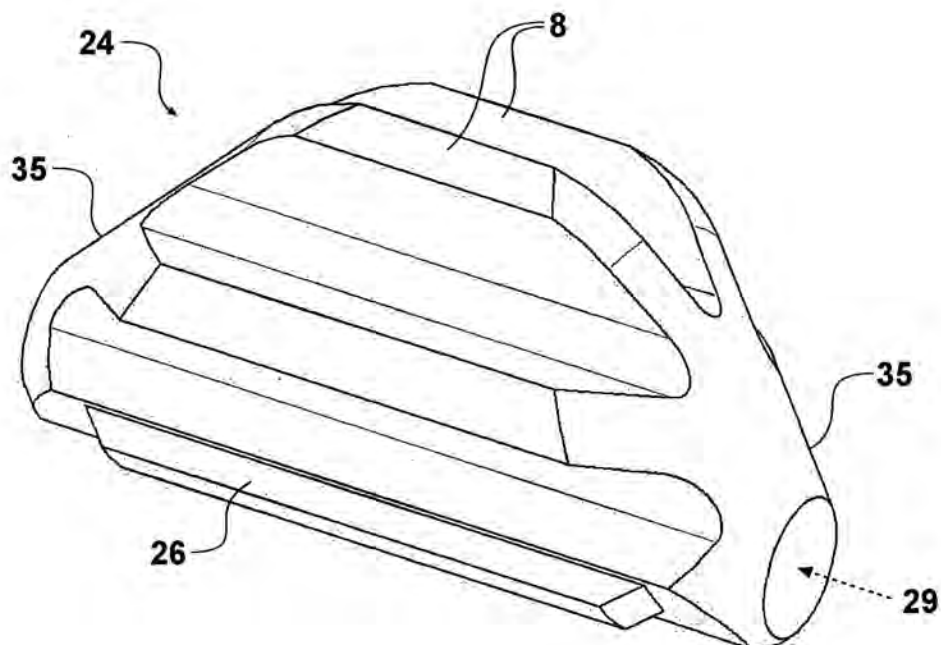


Figure 20

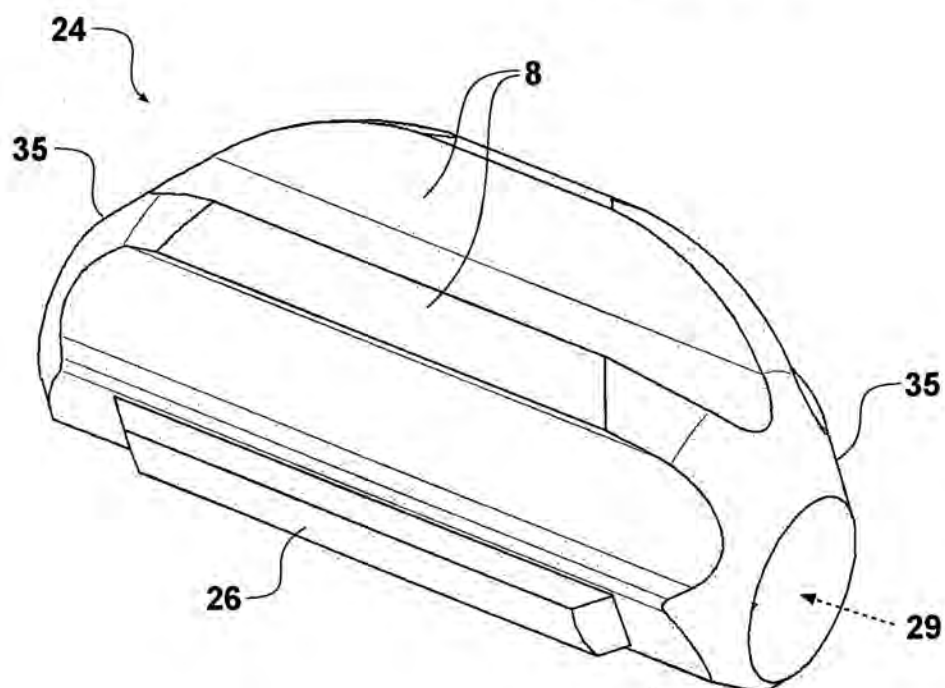


Figure 21

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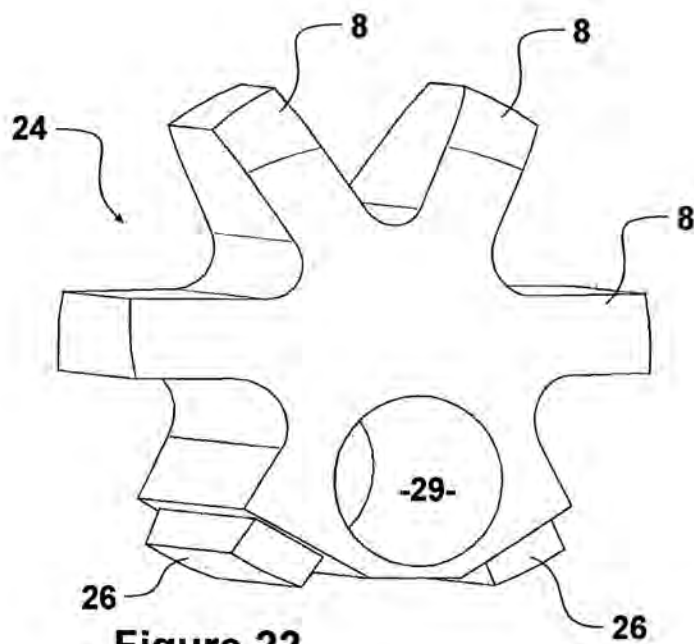


Figure 22

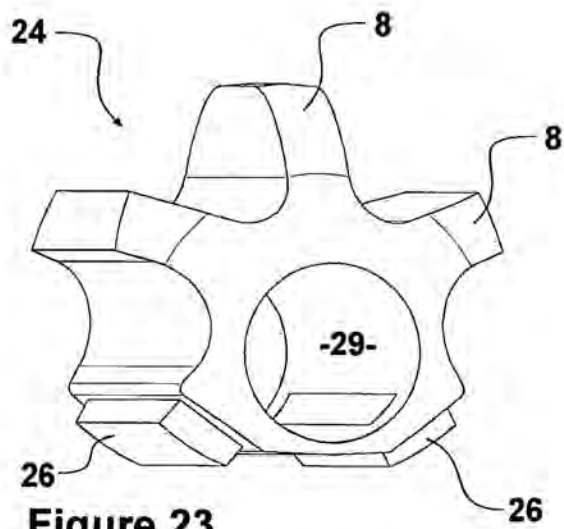


Figure 23

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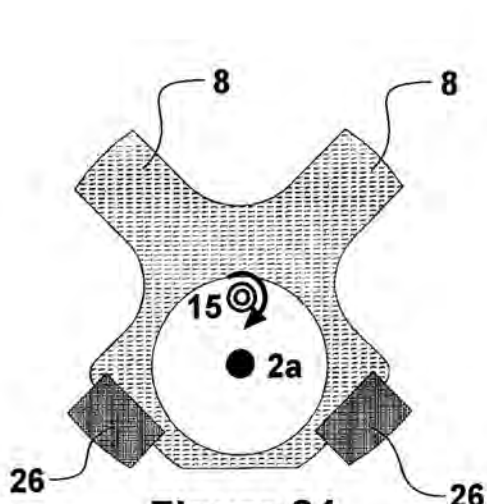


Figure 24

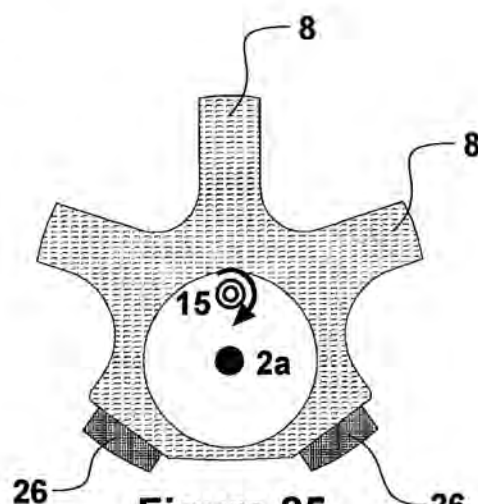


Figure 25

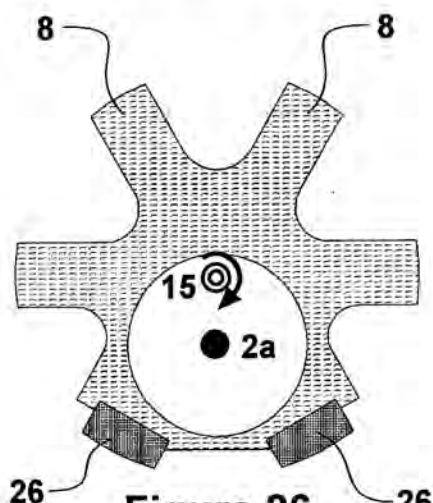


Figure 26

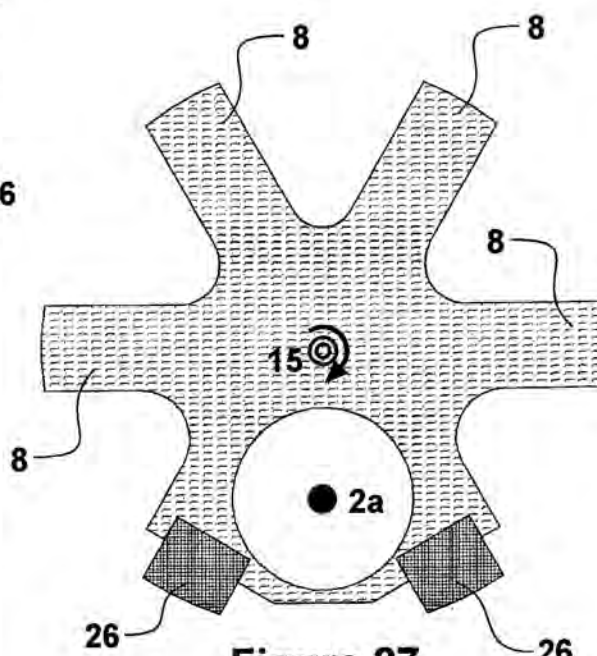
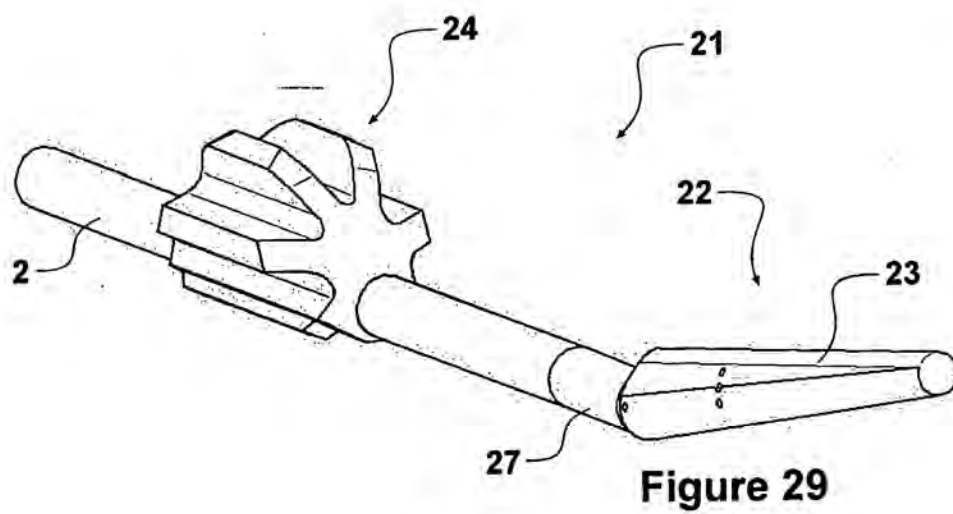
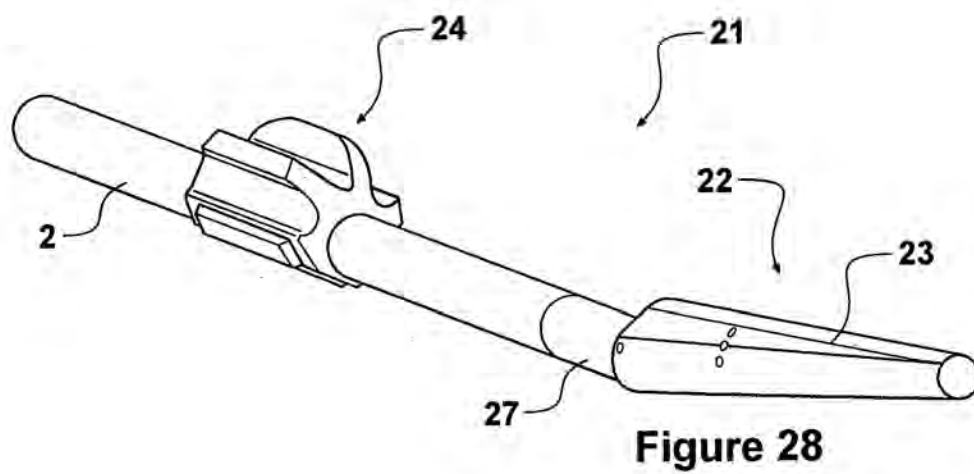


Figure 27

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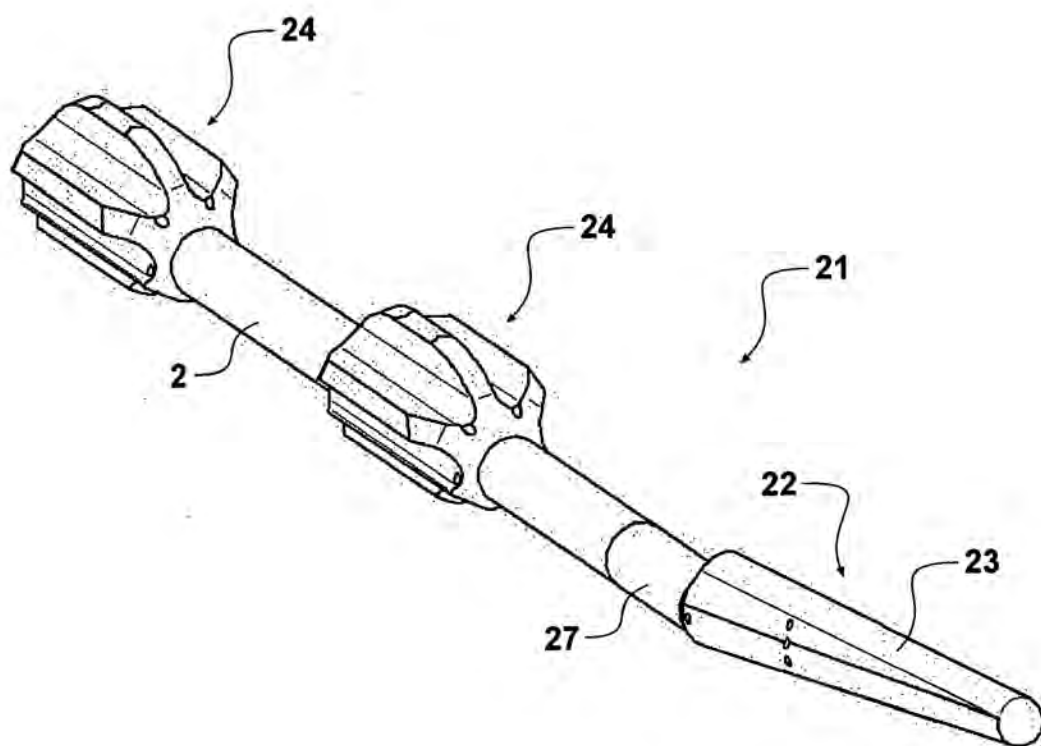
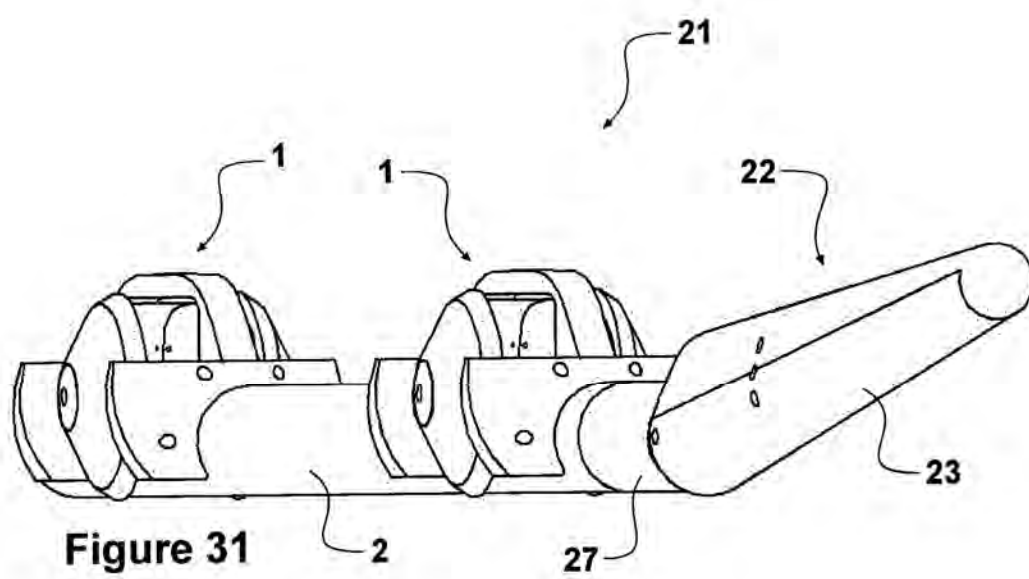


Figure 30

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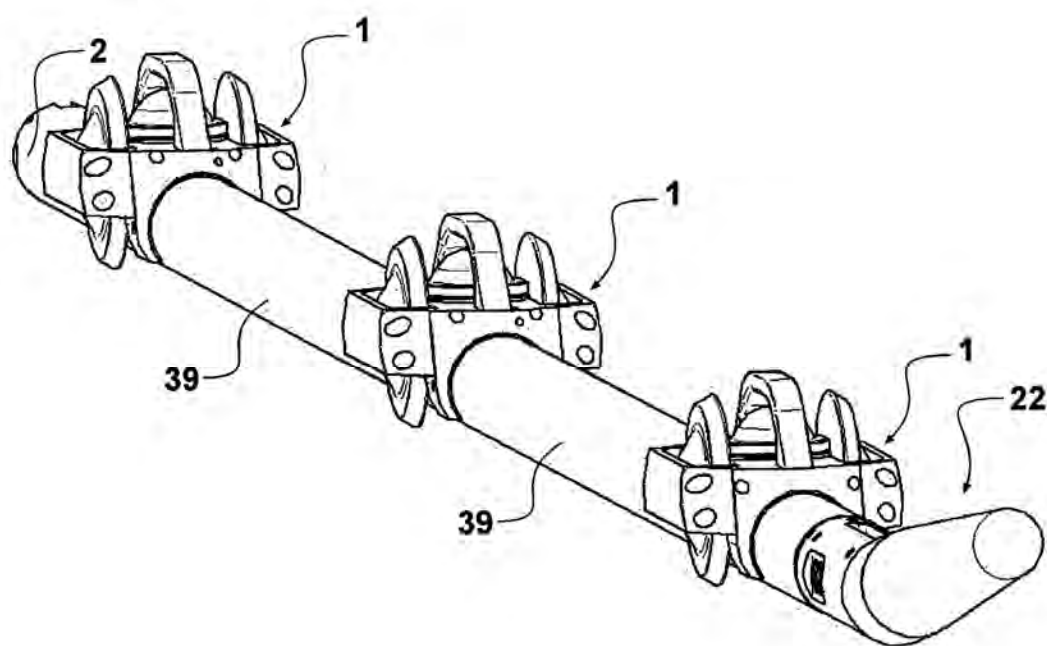


Figure 32

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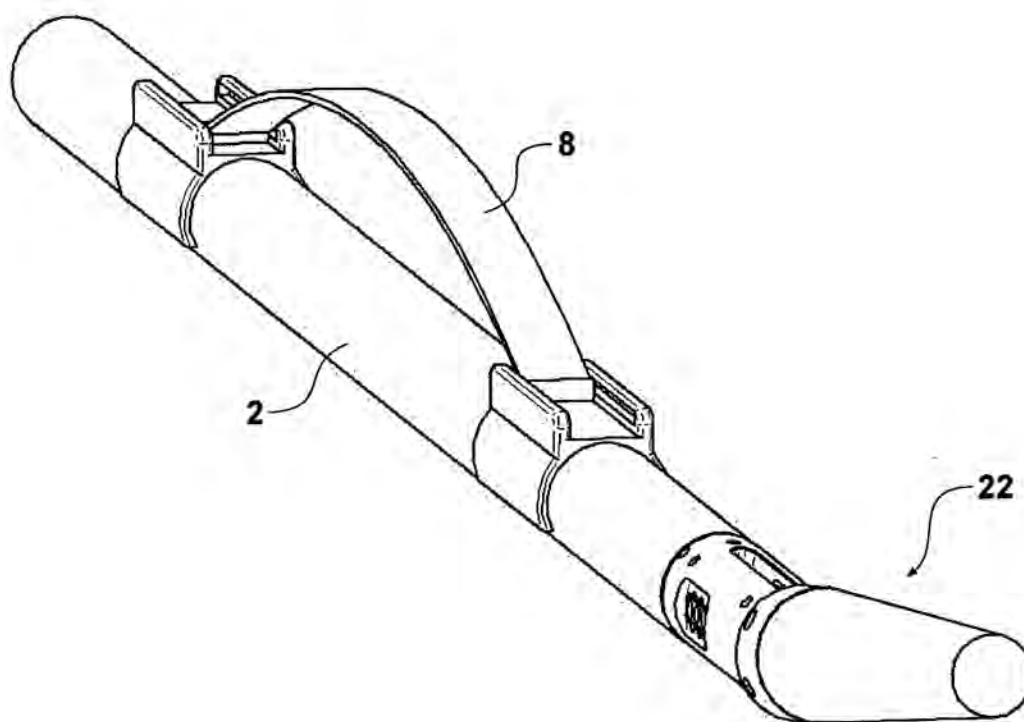


Figure 33

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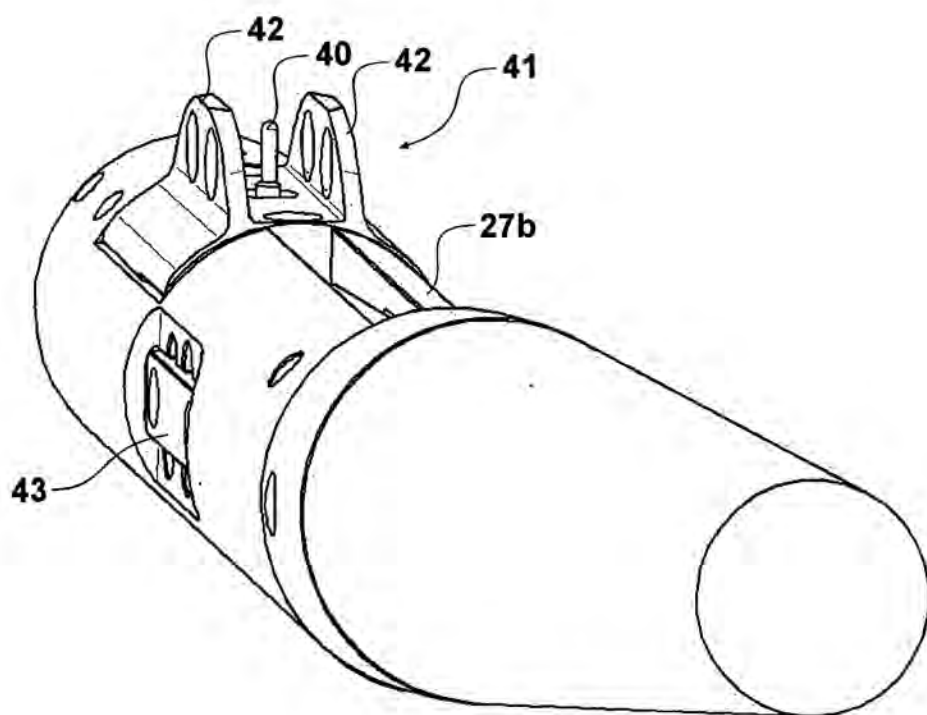


Figure 34

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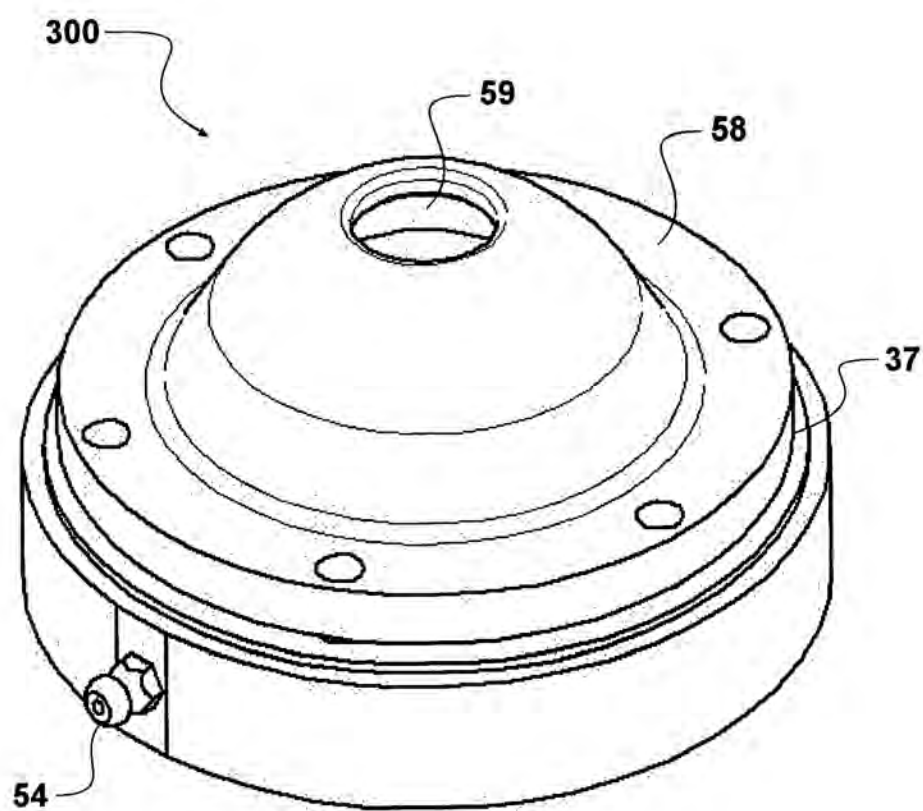
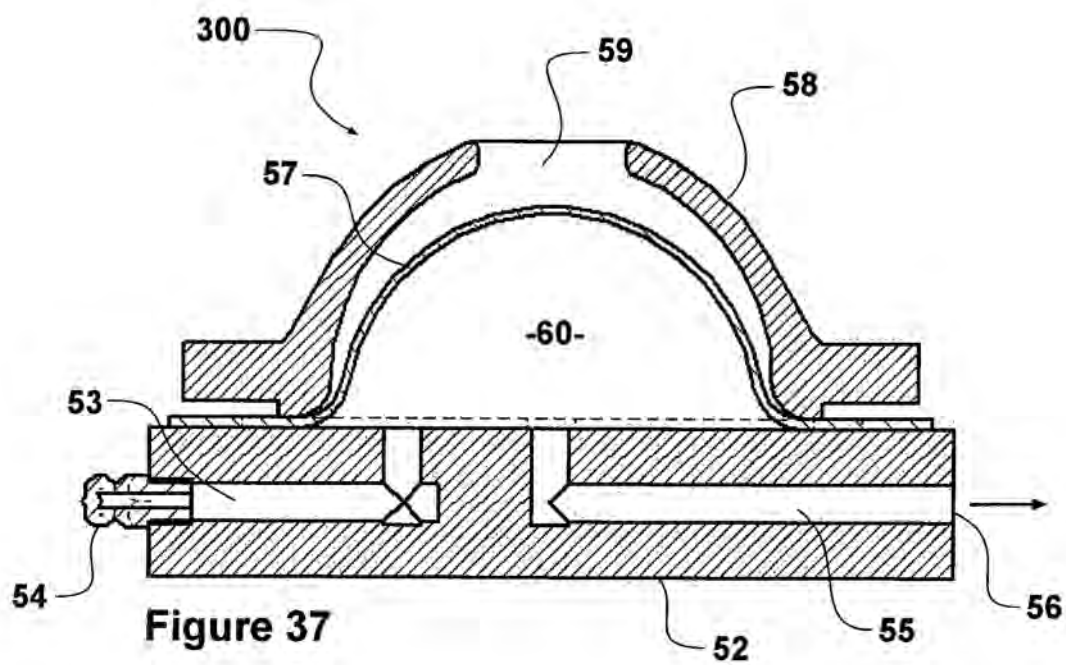
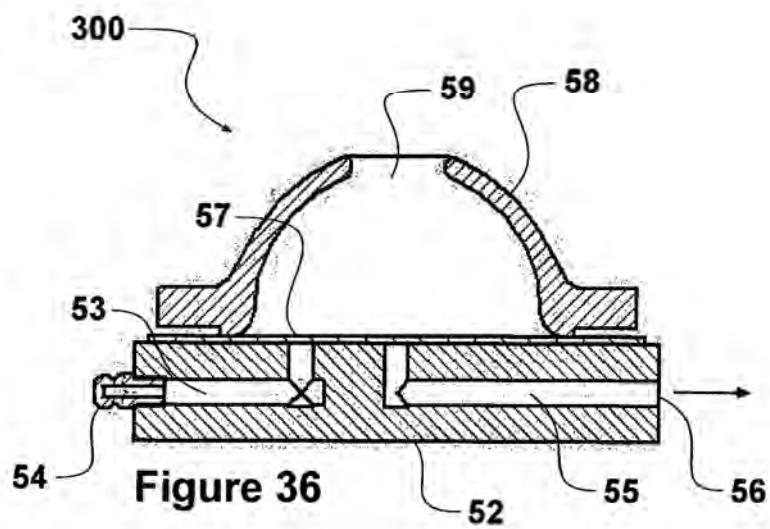
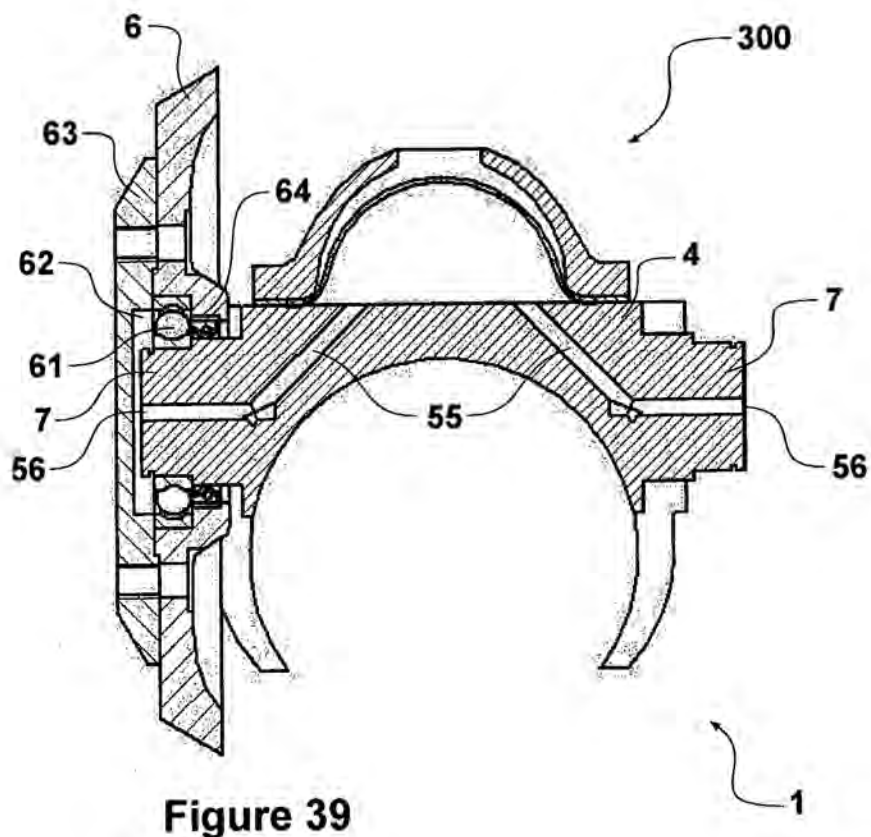
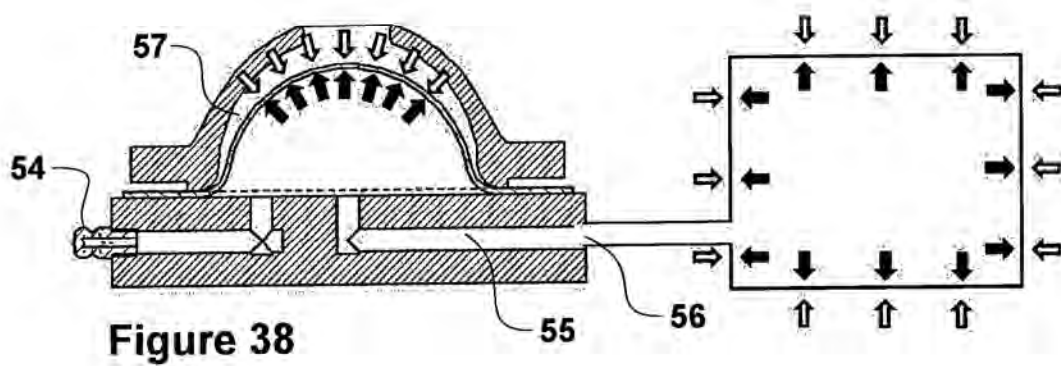


Figure 35

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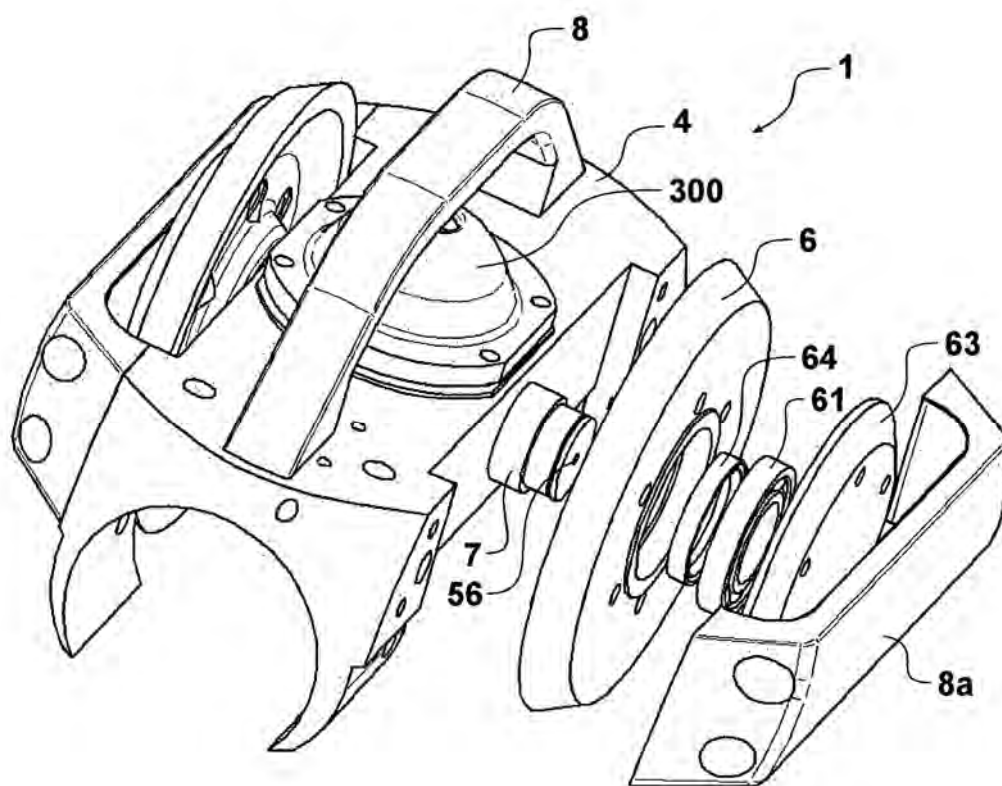


Figure 40

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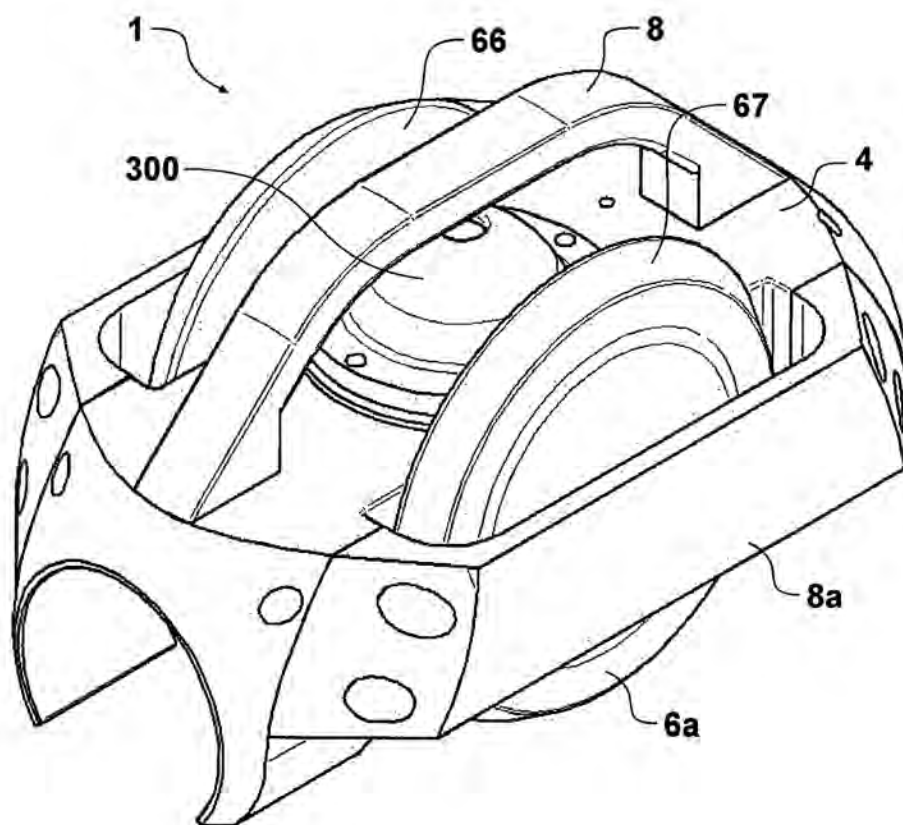


Figure 41

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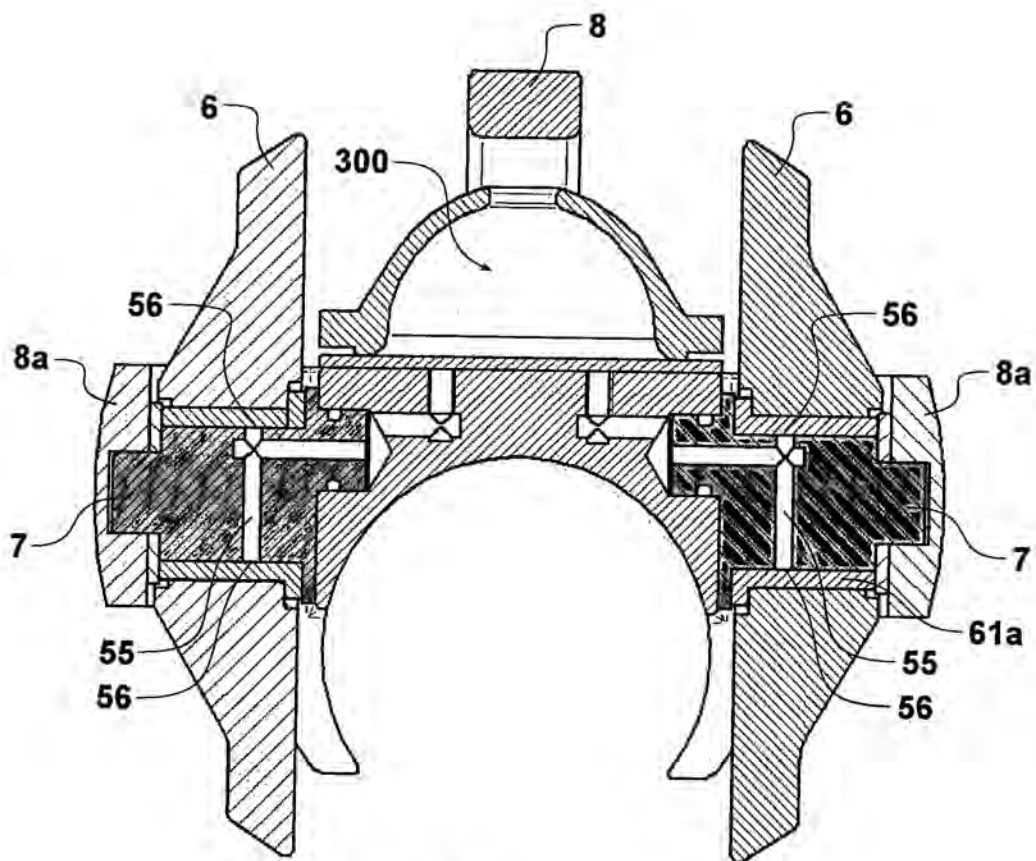


Figure 42

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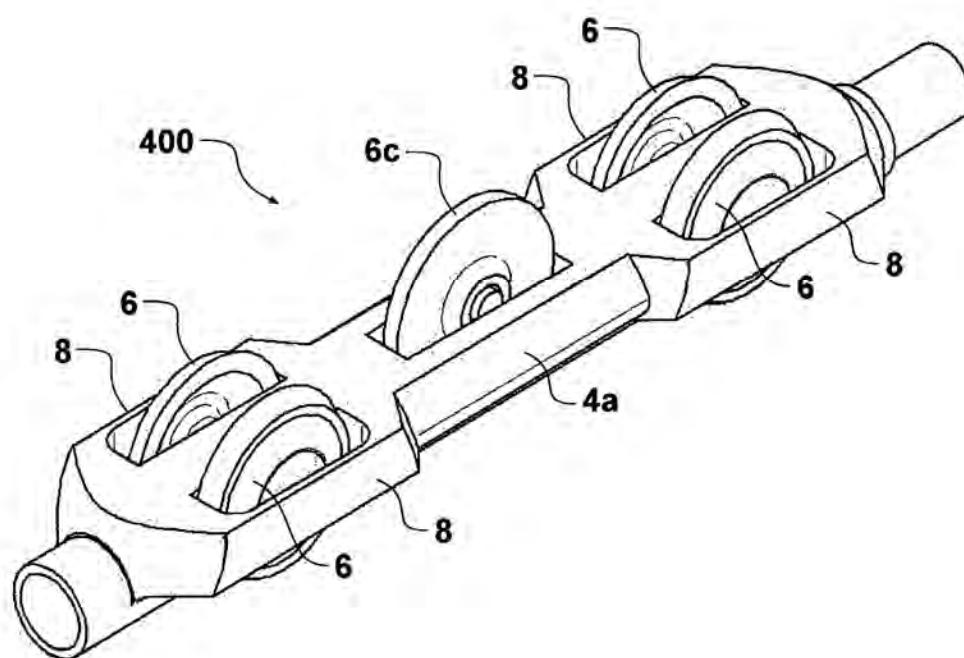


Figure 43

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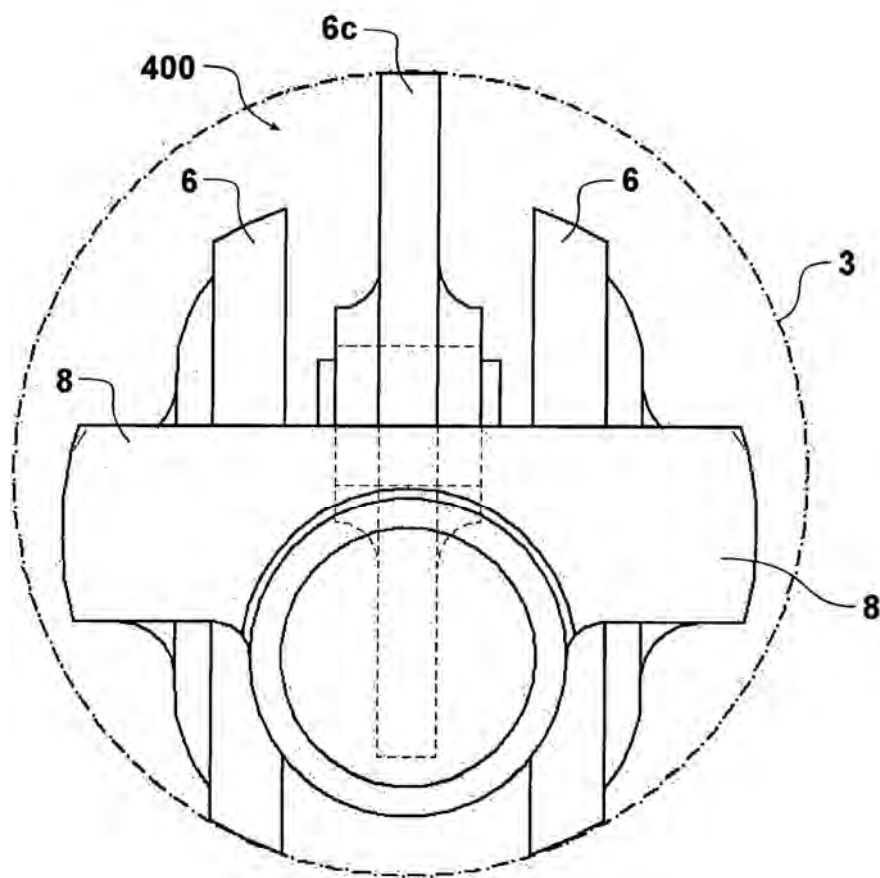


Figure 44

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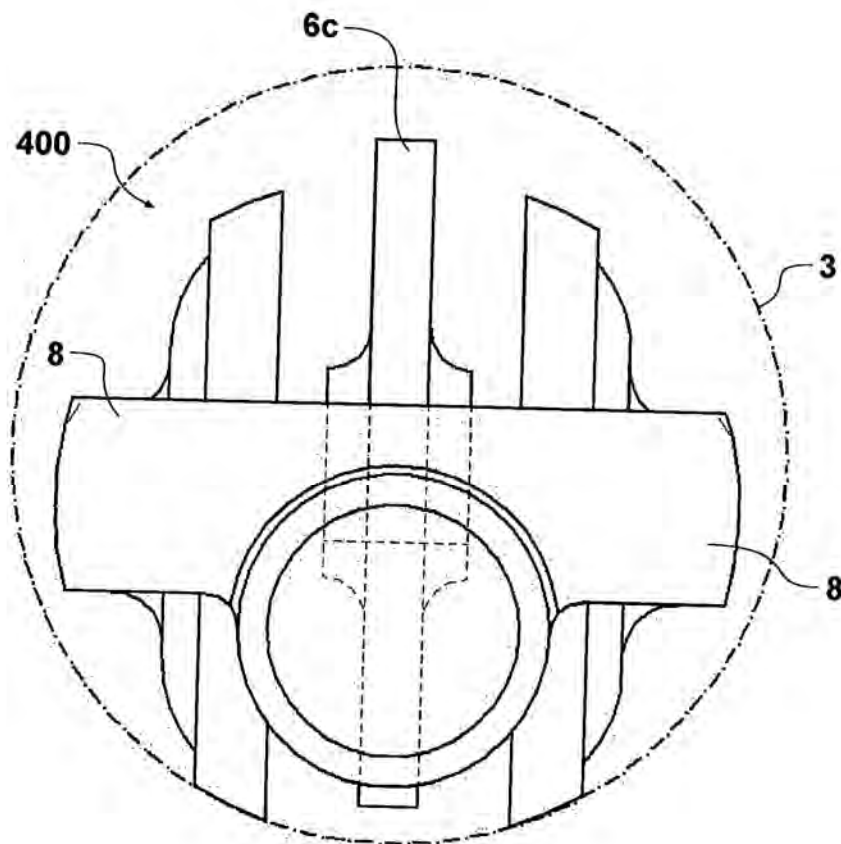


Figure 45

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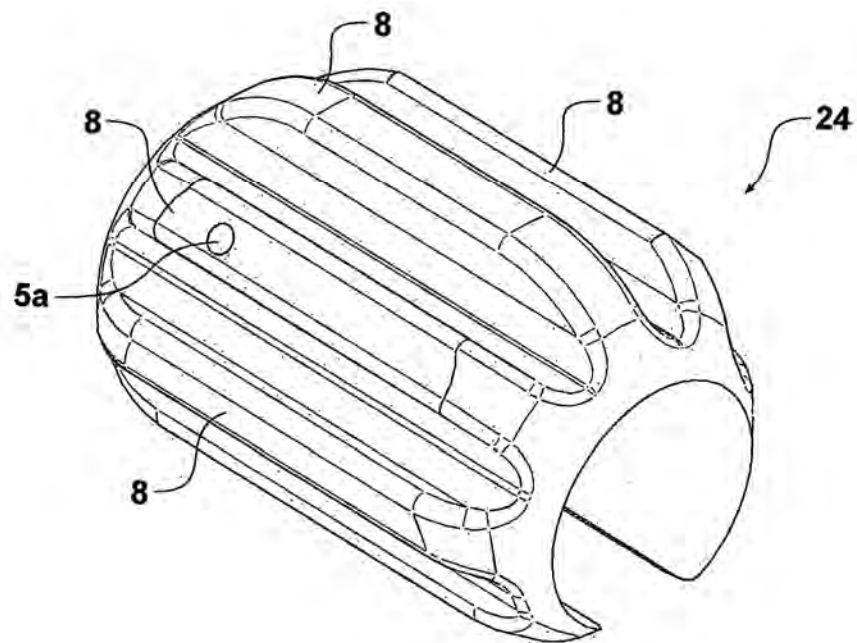


Figure 46

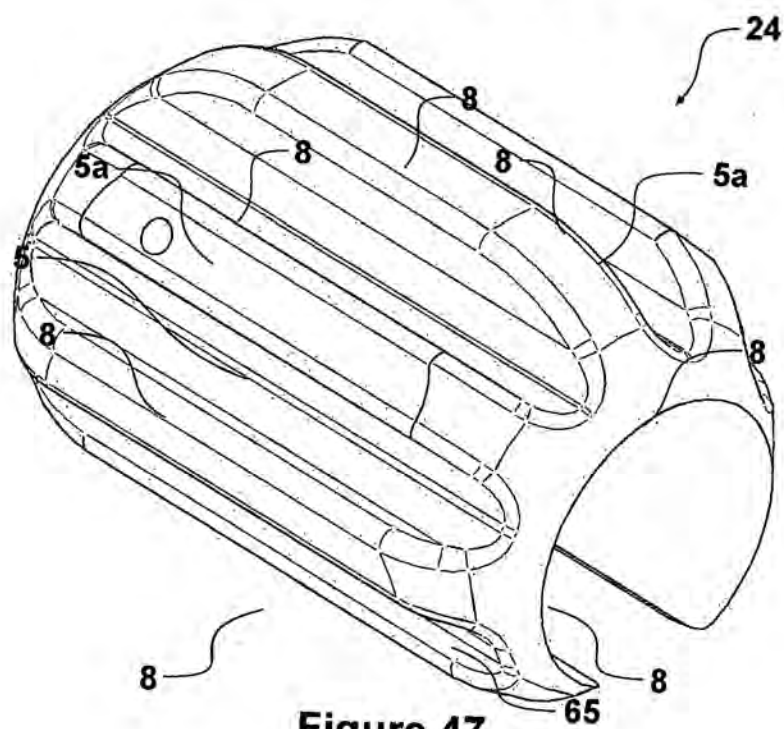


Figure 47

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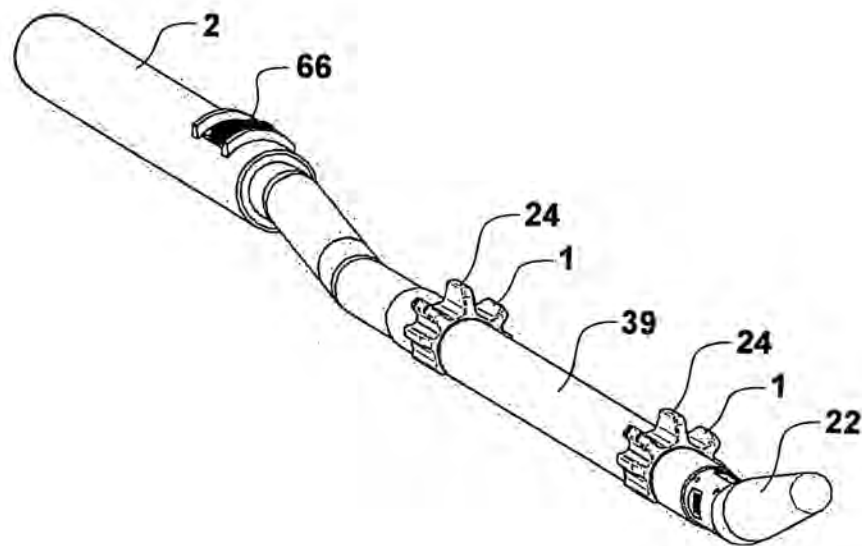


Figure 48

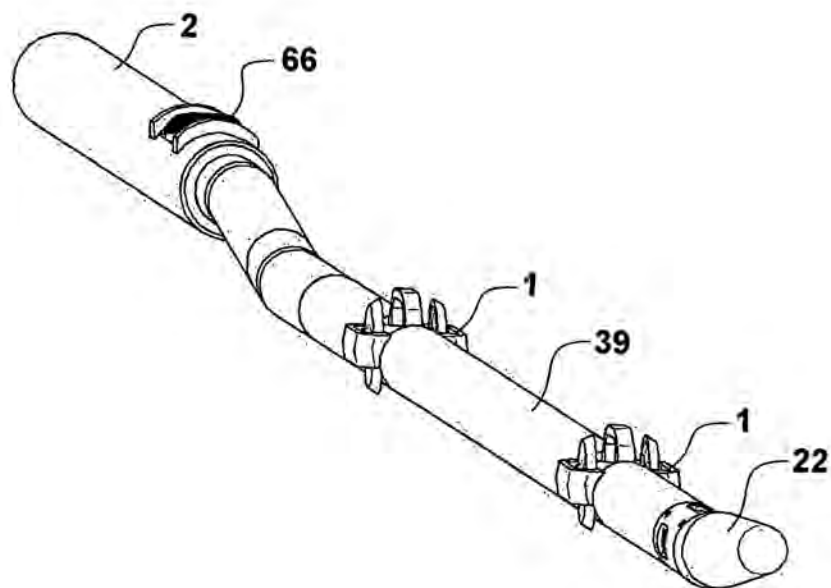


Figure 49